

Forecasting and Managing Correlation Risks

Tim Bollerslev,^a Sophia Zhengzi Li,^b Yushan Tang^{c,*}

^aDuke University and National Bureau of Economic Research, Durham, North Carolina 27708; ^bRutgers Business School, Newark, New Jersey 07102; ^cDishui Lake Advanced Finance Institute, Shanghai University of Finance and Economics, Shanghai 200433, China

*Corresponding author

Contact: boller@duke.edu,  <https://orcid.org/0000-0003-2743-3087> (TB); zhengzi.li@business.rutgers.edu,

 <https://orcid.org/0000-0002-0448-8125> (SZL); tangyushan@sufe.edu.cn,  <https://orcid.org/0000-0001-9615-1804> (YT)

Received: October 28, 2024

Revised: July 10, 2025

Accepted: September 19, 2025

Published Online in *Articles in Advance*:

■■■■■ ■■, 2026

<https://doi.org/10.1287/mnsc.2024.08294>

Copyright: © 2026 INFORMS

Abstract. We propose a novel and easy-to-implement framework for forecasting time-varying correlations based on a large set of salient realized correlation features and the sparsity-encouraging Least Absolute Shrinkage and Selection Operator technique. Considering the universe of S&P 500 stocks, we find that the new approach manifests in statistically superior out-of-sample forecasts compared with commonly used procedures. We further demonstrate how the forecasts translate into significant economic gains in the form of higher pairs trading profits, better equity premium predictions, more accurate portfolio risk targeting, and superior overall risk control and minimization.

History: Accepted by Kay Giesecke, finance.

Funding: S. Z. Li received financial support from the Rutgers Business School Dean's Research Seed Fund.

Supplemental Material: The online appendix and data files are available at <https://doi.org/10.1287/mnsc.2024.08294>.

Keywords: correlation forecasting • high-frequency data • LASSO • risk targeting and control • pairs trading • equity premium prediction

In general, it's more important to know how to diversify well than it is to know how to pick the best assets. —Ray Dalio, July 29, 2022, podcast interview

1. Introduction

Asset pricing research has traditionally focused on the determination of expected returns. Meanwhile, the likely comovements among asset returns arguably plays an equally important, if not greater, role in the construction of well-performing investment portfolios and sound risk management practice. We propose a new and easy-to-implement framework for forecasting time-varying correlations, or “correlation risks,” of stocks by explicitly focusing on out-of-sample prediction rather than in-sample statistical inference. Our approach succinctly combines so-called feature engineering and model fitting into a coherent framework. In the feature engineering step, we consider a number of variables that have previously been used in the literature, together with several new purposely designed features. In the model fitting step, we deliberately select the most useful features through the use of sparsity-encouraging estimation procedures. Our empirical results, based on intraday high-frequency data for a large sample of S&P 500 stocks spanning several decades, demonstrate highly statistically significant improvements in the accuracy of the out-of-sample forecasts compared with commonly used popular benchmark models. These

statistical improvements also translate into substantial economic gains in portfolio construction, risk control, and return prediction, further underscoring the practical value of the new procedures.

Guided by the ideas of Kelly et al. (2024) among others, encouraging the use of *all* plausibly relevant predictors to improve the performance of machine learning-based procedures, we consider several alternative feature sets, each of which might ostensibly contain useful predictive information.¹ Consistent with well-documented autoregressive dynamic dependencies in correlations, the first feature set naturally consists of lagged daily, weekly, and monthly realized correlation measures, together with their semicorrelation extensions, as formally defined below. The second feature set captures the common factor structures of stock returns by projecting the realized covariance matrix onto the space spanned by observable firm characteristics, following the approach of Fama and French (2020) and Li et al. (2025). This process effectively isolates the component of return comovement attributable to systematic variation, thereby combining high-frequency realized covariance estimates with low-frequency cross-sectional firm characteristics. Our third feature set considers a smoothed version of the realized correlations that explicitly incorporates sector information into the more traditional measures.

Having defined the different feature sets, we next develop a correlation forecast model by simultaneously

combining all features into a simple linear model. We consider both ordinary least squares (OLS) and the sparsity-encouraging least absolute shrinkage and selection operator (LASSO) (Tibshirani 1996) for fitting the models. We find that jointly fitting all features via OLS typically outperforms the widely used heterogeneous autoregressive (HAR) forecasting model (Corsi 2009).² However, sparsely fitting the features via LASSO results in additional significant improvements over all the OLS-based models. One key step of the LASSO fitting underlying these results consists of feature selection via model tuning on validation samples, in which we chronologically split the sample into separate training and validation sets, used for estimation and model selection, respectively. We do so on a rolling-window basis that updates annually. This approach in turn reveals interesting time-varying patterns in the selected features in concert with varying economic and financial market conditions.

To more concretely evaluate the economic significance of the new correlation forecasting models, we consider four distinct practical applications. In the first application, we study the augmented pairs trading strategy of Chen et al. (2019). We show that our new correlation forecasting models significantly improve the equal-weighted (value-weighted) strategy returns to 9.28% (8.83%) per annum, compared with 3.63% (6.14%) based on HAR-model predictions. Our second application is motivated by Pollet and Wilson (2010), who show that the average correlation among individual stock returns can help predict aggregate market returns. We find that the predicted correlations from our new models yield substantially stronger predictive power, with adjusted monthly R^2 values of 1.91% (1.92%) for the equal-weighted (value-weighted) market index, compared with 0.74% (0.71%) for the traditional HAR-based forecasts—results that are comparable to some of the best-performing market predictors. Our third application evaluates the risk of various long-short portfolio strategies and style tilts by comparing realized portfolio risks with their forecasted counterparts. We again find that the LASSO-based forecasts significantly improve the alignment between predicted and realized risks, with the average risk-targeting ratio across all strategies equal to 1.02, versus 0.83 for the HAR-based forecasts. Our fourth and final application considers the construction of Global Minimum Variance (GMV) portfolios. We show that the new forecasting models achieve the lowest realized risks for the GMV portfolios. Moreover, using the utility framework of Fleming et al. (2003), we find that a risk-averse beta-neutral GMV investor would be willing to sacrifice up to 6.07% in annual return to switch from the simple HAR-based forecasts to our new LASSO-based forecasts.

To assess the robustness and sensitivity of our main results, we further split the full sample into smaller

subsamples. Although the overall quality of model fits varies somewhat across these subsamples, the LASSO-based models consistently deliver superior out-of-sample forecasts, including during the recent COVID-19 period. We also examine whether the inclusion of additional economically motivated features, derived from various notions of firm linkages, are able to further improve on the predictive accuracy. We find that none of these added features meaningfully enhances the out-of-sample forecast performance relative to the forecasts generated from our three specifically engineered feature sets. In addition, we explore a series of alternative model fitting procedures drawn from the machine learning (ML) literature, including Ridge Regression, Elastic Net, Principal Component Regression, and Feed-Forward Neural Networks. Again, none of these alternative fitting procedures improves significantly on the performance of our simple-to-implement LASSO-based forecasts. Finally, we also expand the stock universe to include all historical constituents of the S&P 500 index to alleviate concerns about potential survivorship bias. Our framework continues to perform well in this extended unbalanced sample.

We make a number of contributions to the literature. First, our construction of new feature sets specifically designed for correlation forecasting, which succinctly combine information from high-frequency market data and low-frequency fundamental data, is decidedly new. Second, our use of sparsity-encouraging fitting techniques to robustly exploit big data with many observations and features for more accurate large-scale correlation forecasting is similarly new to the literature. Third, our illustration of the strong economic gains afforded in a wide range of practical applications adds importantly to our understanding of the new procedures and further underscores the value of better risk forecasts more generally.

Our paper is related to several strands of literature—most closely, perhaps, to the large financial econometrics literature, dating back to Engle (2002) and Tse and Tsui (2002), on modeling time-varying conditional correlations using generalized autoregressive conditional heteroskedasticity (GARCH)-type models. Extending these earlier works, the parametric models proposed by Cappiello et al. (2006) and Audrino and Trojani (2011) explicitly allow for asymmetric dynamic dependencies in the conditional correlations. More recently, Engle et al. (2019) have suggested combining the original dynamic conditional correlation (DCC) model of Engle (2002) with shrinkage techniques derived from random matrix theory to allow for improved estimation in large dimensions, while Johansson et al. (2023) propose a simple extension of DCC-type models based on iterated estimation. These multivariate GARCH-type models have also subsequently been extended to incorporate realized variation measures in the

construction of more accurate forecasts; see, for example, Noureldin et al. (2012), along with the more recent paper by Bollerslev et al. (2020a) and the many other studies discussed therein. Instead, we focus directly on forecasting the realized correlation measures constructed from intraday data. The idea of directly modeling and forecasting realized variation measures dates back to Andersen et al. (2003), and it has been explored extensively in the literature since then, albeit mostly in the context of forecasting realized variances; see also the recent discussion and literature review in Bollerslev (2022). Our paper importantly extends this existing literature by allowing for more flexible dynamics and a much wider set of predictor variables than has hitherto been considered.

The paper also adds to the burgeoning literature on the use of ML techniques for financial decision making. In addition to the already large literature devoted to return prediction, as exemplified by Rapach et al. (2013), Gu et al. (2020), Li and Rossi (2025), Bali et al. (2022), Chen et al. (2024), Kaniel et al. (2023), Bali et al. (2023), and Lopez-Lira and Roussanov (2023), our paper is most closely related to the recent literature on applying ML techniques for the purpose of risk management. The studies by Audrino and Knaus (2016), Bucci (2020), Christensen et al. (2023), and Li and Tang (2025), in particular, all rely on ML learning algorithms for univariate volatility forecasting, while Bollerslev et al. (2022b) do so for covariance matrix forecasting. Bollerslev et al. (2022a) similarly adapt ideas from ML to cluster stocks into groups based on their realized risk characteristics.

Our new approach differs from these existing studies in two important dimensions. First, we focus on specifically designed and economically motivated feature sets, along with a deliberately chosen sparsity-inducing fitting technique to build reliable forecasting models. Second, rather than emphasizing purely statistical assessments of the correlation predictions, we further demonstrate the economic value of our procedures across a range of practical financial applications. Another strand of recent research has similarly sought to enhance the economic interpretability of ML models through the use of flexible tree-based algorithms in the context of asset pricing and return predictability, as exemplified by the work of Gu et al. (2020), Cong et al. (2024), and Cong et al. (2025). It is, of course, possible that the integration of these or other more complex nonlinear formulations and machine learning techniques may uncover additional structures beyond the linear feature dependencies underlying our LASSO-based forecasts. However, our robustness analysis also suggests that it is difficult to do so.

The paper is also related to the large finance literature on stronger comovements among certain types of stocks, including S&P 500 index constituents (Barberis et al. 2005), firms with similar institutional ownership

(Pindyck and Rotemberg 1993), firms with headquarters in the same geographical location (Pirinsky and Wang 2006), and firms with similar analyst coverage (Muslu et al. 2014, Hameed et al. 2015, Israelsen 2016), to name but a few. In contrast to all these studies, however, which primarily focus on causal relations between firm linkages and asset price movements, we focus explicitly on the prediction of future stock return correlations. In so doing, we show empirically that these previously established causal firm-level linkage measures provide limited predictive power over and above that afforded by the new specifically designed features proposed here.

The paper is organized as follows. Section 2 discusses the data and the different features underlying our empirical analyses. Section 3 summarizes the details of the estimation methodology that we rely on. Section 4 compares the statistical out-of-sample forecasting performance of the new models to existing procedures. Section 5 highlights the economic significance of our new forecasting models by considering their uses in four distinct practical applications. Section 6 summarizes the results from additional robustness checks and empirical analyses. Section 7 concludes. Further details regarding various modeling choices along with data construction and additional results are deferred to an [appendix](#) and an Online Appendix.

2. Data and Variables

2.1. Data

We consider the universe of stocks that were ever constituents of the S&P 500 index and have full historical data from the New York Stock Exchange (NYSE) Trade and Quote (TAQ) database between January 2000 and December 2020.³ We further require the stocks to be listed on the NYSE, National Association of Securities Dealers Automated Quotations (NASDAQ), and the American Stock Exchange (AMEX) with share codes of 10 or 11, prices between \$1 and \$1,000, and at least 100 daily trades. To alleviate concerns about bid-ask bounce effects (Roll 1984) that, together with nonsynchronous prices, are well-known to induce downward biases in correlation estimation at ultrahigh frequencies (Epps 1979), we rely on 15-minute intraday returns based on “coarsely” sampled midquote prices.⁴ Some of our realized features require a sufficiently long return history for their construction. To accommodate this, we use the data between 2000 and 2002 to compute the initial observations for these variables, fixing January 2003 as the common start date for our full-sample analysis. All in all, this leaves us with a final stock sample consisting of 417 unique S&P 500 stocks with full historical data for all features and response variables over the period from January 2003 to December 2020.

We also consider two additional sets of stock-level data for our feature construction and performance evaluation. The first set consists of 15 representative anomalies, including the 11 mispricing anomalies from Stambaugh et al. (2012), together with the traditional CAPM Beta, Size, Book-to-Market, and Reversal measures. We further complement this first set of anomalies with another group of 15 firm-level characteristics that have also been widely studied in the literature, including measures related to firm earnings, growth, and momentum. All of the anomalies are constructed following Green et al. (2013) and Chen and Zimmermann (2022) based on data from Compustat and CRSP. More detailed discussion and summary statistics are provided in the Online Appendix.

2.2. Response Variable

Our main research objective centers on the development of better predictive models for monthly correlations. Accordingly, we rely on measures of monthly realized correlations as our response variable. Our use of the realized correlation measures may be formally justified by the theory of quadratic variation. In particular, assuming that the joint dynamics of the stocks adhere to some underlying arbitrage-free Itô semimartingale vector process (Back 2010), the true covariance matrix over a given time interval, like a day or a month, may then be consistently estimated by the summation of ever-finer-sampled within-interval cross-products of returns (Andersen et al. 2003, Barndorff-Nielsen and Shephard 2004).

To set up the requisite notation, let $p_{i,d}$ denote the natural logarithm of stock i 's price on day d . Assume that intraday prices are observed at $n + 1$ equally spaced time intervals from day d to day $d + 1$ —say, $p_{i,d}, p_{i,d+1/n}, \dots, p_{i,d+1}$ —with the corresponding returns denoted by $r_{i,d-1+k/n} \equiv p_{i,d-1+k/n} - p_{i,d-1+(k-1)/n}$ for $k = 1, \dots, n$. The annualized daily realized variance for stock i on day d and the realized covariance between stocks i and j on day d are then simply constructed as:

$$\begin{aligned} RV_i^d &\equiv 252 \times \sum_{k=1}^n r_{i,d-1+k/n}^2 \\ RCov_{ij}^d &\equiv 252 \times \sum_{k=1}^n r_{i,d-1+k/n} \cdot r_{j,d-1+k/n}. \end{aligned} \quad (1)$$

Although the theory formally underlying consistency of the realized measures calls for the use of increasingly finer sample intraday returns, or $n \rightarrow \infty$, as previously noted to help mitigate the effects of nonsynchronous trading and other market microstructure effects that might bias the estimates, in the results reported on below, we deliberately rely on a “coarse” 15-minute sampling frequency, or $n = 27$.⁵ In addition, to obtain an estimate of the variation for the entire day, we

follow common practice in the literature (see, e.g., Hansen and Lunde 2005) and include the overnight returns in the calculations of the full-day RV^d 's and $RCov^d$'s. We obtain longer-run weekly and monthly measures, denoted by RV^w 's, $RCov^w$'s, RV^m 's, and $RCov^m$'s in the sequel, by averaging the daily measures over one week and one month, respectively.

In line with the majority of the existing asset pricing literature, we will primarily focus on a one-month forecast horizon. Accordingly, to facilitate the notation, we will often drop the m superscript and use the time index t to refer to a month. Correspondingly, we will denote the monthly realized correlation matrix by RC_t , which is implicitly defined as follows:⁶

$$RCov_t = \sqrt{RV_t} \cdot RC_t \cdot \sqrt{RV_t}, \quad (2)$$

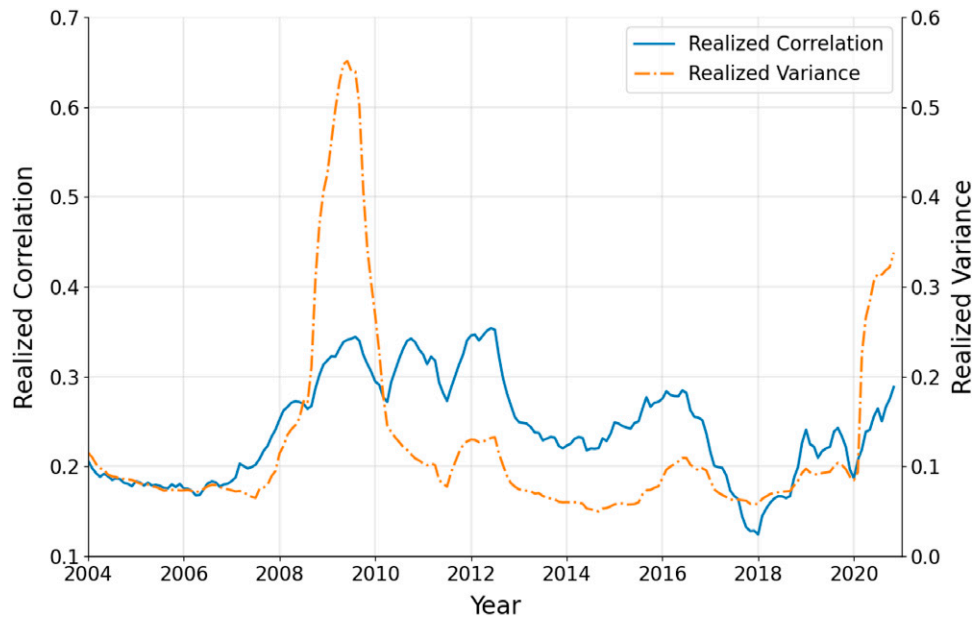
where RV_t refers to the diagonal matrix with the month- t realized variances along the diagonal, and $RCov_t$ denotes the month- t realized covariance matrix. There is, of course, already a vast literature on modeling and forecasting realized variances; see, for example, the recent discussions and literature reviews in Bollerslev et al. (2018b), Christensen et al. (2023), and Li and Tang (2025). Instead, we focus exclusively on forecasting realized correlations and their practical applications.

Meanwhile, it is well-established that RV_t and RC_t tend to exhibit quite different dynamic dependencies (see, e.g., the discussion in Oh and Patton 2016). To illustrate, Figure 1 plots the 12-month moving average of the cross-sectional means of the monthly realized correlations and realized variances over the full sample period. Even though the time series of realized correlations appear relatively more stable, and clearly more so than the realized variances, the series still reveals nontrivial temporal variation in the cross-sectional average monthly realized correlations, with notably higher values in the aftermath of the 2008 financial crisis, as well as the recent COVID-19 period. Figure 2, which displays the unconditional distribution of all the monthly realized correlations, further highlights the dispersion in the individual monthly stock-pair correlations across time and stocks. We turn next to a discussion of the different features that we use in the construction of the new and improved forecasting models for monthly realized correlations.

2.3. Features

We begin our analysis by constructing a series of input features that potentially contain predictive information for the one-month-ahead realized correlations. We consider three separate sets of features: (1) features inspired by HAR-type models ($RC^d, RC^w, RC^m, RC^{d-}, RC^{w-}, RC^{m-}$), (2) features constructed from common factors (FRC^d, FRC^w, FRC^m), and (3) exponentially weighted realized features ($ExpRC^d, ExpRC^w, ExpRC^m$,

Figure 1. (Color online) Monthly Realized Correlations and Variances



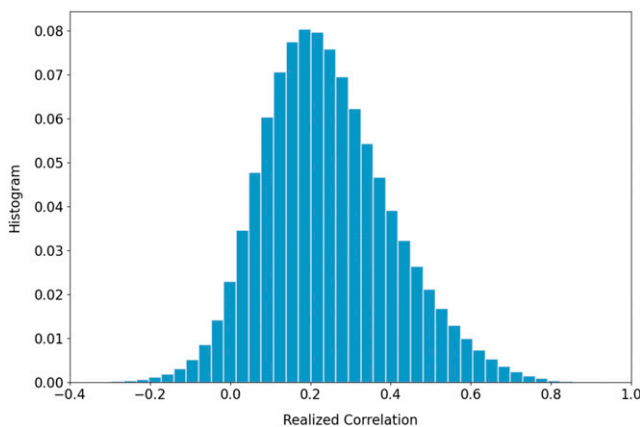
Note. The figure plots the 12-month moving average of the cross-sectional means of the monthly realized correlations and realized variances for the S&P 500 stocks in our sample.

$ExpRC^q$, $ExpRC^{d-}$, $ExpRC^{w-}$, $ExpRC^{m-}$, $ExpRC^{q-}$, $ExpScRC^d$, $ExpScRC^w$, $ExpScRC^m$, $ExpScRC^q$, $ExpScRC^{d-}$, $ExpScRC^{w-}$, $ExpScRC^{m-}$, $ExpScRC^{q-}$).⁷ Definitions of the features are detailed in the [appendix](#).

The three different feature sets discussed above comprise a total of 25 unique correlation predictors for each stock pair.⁸ Table 1 reports a series of descriptive statistics for the features. There are several noticeable patterns. First, the sample means of all the features are positive, indicating that stocks tend to move in the same direction, on average. Second, the means of the factor-driven realized features (FRCs) are only slightly below those measured from the returns (RCs), suggesting that

much of the comovement among the stocks can indeed be accounted for by common factors. Third, the negative realized semifeatures (RC^- 's, $ExpRC^-$'s, and $ExpScRC^-$'s) tend to have higher means than their unsigned counterparts (RC s, $ExpRC$ s, and $ExpScRC$ s), consistent with the idea of stronger asset comovements during market downturns. Fourth, the exponential realized features ($ExpRC$ s) generally have lower standard deviations than the more traditional calendar-based realized features (RC s) due to the greater number of lagged daily realized correlations used in the smoothing. Lastly, the sector-specific exponential features ($ExpScRC$'s and $ExpScRC^-$'s) naturally have the lowest overall sample means, as they are fixed at zero for any stock pair belonging to different sectors.

Figure 2. (Color online) Monthly Realized Correlations



Note. The figure displays the histogram of monthly realized correlations for the S&P 500 stocks in our sample.

3. Estimation Methodology

Our competing forecasting models are based on linear specifications and two fitting procedures: Ordinary Least Squares and the Least Absolute Shrinkage and Selection Operator. The former entails a predefined set of predictors, whereas the latter performs variable selections to reduce the dimension of the feature sets as part of the model estimation.

3.1. Training and Validation

In parallel to many machine learning algorithms, LASSO requires a validation set for tuning its hyperparameter(s). In the case of LASSO, this amounts to a single penalty parameter on the sum of the absolute slope coefficients, which succinctly controls the number

Table 1. Descriptive Statistics for Realized Correlation Features

Variable	Mean	Std	Skewness	Kurtosis	Min	P1	P25	Median	P75	P99	Max	AR(1)	AR(5)	AR(21)	AR(63)
RC^d	0.26	0.30	-0.33	-0.21	-0.98	-0.49	0.06	0.28	0.48	0.85	1.00	0.09	0.03	0.03	-0.04
RC^w	0.25	0.21	0.06	0.02	-0.93	-0.22	0.11	0.25	0.39	0.74	1.00	0.20	0.09	0.10	-0.05
RC^m	0.24	0.16	0.36	0.25	-0.89	-0.10	0.13	0.23	0.34	0.66	0.98	0.40	0.18	0.11	-0.04
RC^{d-}	0.46	0.22	0.07	-0.83	0.00	0.04	0.29	0.46	0.63	0.91	1.00	0.08	0.03	0.02	-0.03
RC^{w-}	0.44	0.16	0.21	-0.32	0.00	0.11	0.32	0.43	0.55	0.81	1.00	0.16	0.07	0.08	-0.03
RC^{m-}	0.41	0.13	0.31	0.07	0.01	0.14	0.33	0.41	0.50	0.74	0.98	0.28	0.14	0.11	-0.03
FRC^d	0.24	0.22	0.91	1.76	-1.00	-0.14	0.09	0.20	0.35	0.98	1.00	0.32	0.17	0.09	-0.08
FRC^w	0.22	0.17	0.97	1.51	-1.00	-0.07	0.10	0.19	0.31	0.75	1.00	0.46	0.26	0.14	-0.10
FRC^m	0.21	0.15	0.94	1.53	-1.00	-0.05	0.10	0.18	0.29	0.66	1.00	0.60	0.33	0.14	-0.09
$ExpRC^d$	0.25	0.22	-0.03	0.04	-0.95	-0.26	0.10	0.25	0.40	0.75	0.99	0.17	0.07	0.08	-0.05
$ExpRC^w$	0.24	0.17	0.34	0.31	-0.88	-0.11	0.12	0.23	0.34	0.67	0.97	0.40	0.15	0.13	-0.05
$ExpRC^m$	0.24	0.14	0.52	0.43	-0.69	-0.03	0.14	0.22	0.32	0.61	0.93	0.79	0.35	0.12	-0.15
$ExpRC^q$	0.24	0.12	0.54	0.37	-0.46	0.00	0.16	0.23	0.32	0.58	0.93	0.93	0.64	0.17	-0.30
$ExpRC^{d-}$	0.44	0.17	0.21	-0.41	0.00	0.11	0.32	0.43	0.55	0.83	0.99	0.14	0.07	0.07	-0.03
$ExpRC^{w-}$	0.41	0.13	0.32	0.08	0.01	0.13	0.32	0.41	0.50	0.75	0.98	0.33	0.13	0.12	-0.04
$ExpRC^{m-}$	0.40	0.11	0.38	0.25	0.03	0.17	0.32	0.39	0.47	0.70	0.96	0.75	0.30	0.11	-0.12
$ExpRC^{q-}$	0.40	0.10	0.42	0.22	0.04	0.19	0.33	0.39	0.46	0.66	0.95	0.91	0.61	0.16	-0.25
$ExpScRC^d$	0.04	0.12	2.98	8.30	0.00	0.00	0.00	0.00	0.00	0.54	0.82	0.40	0.18	0.21	-0.10
$ExpScRC^w$	0.04	0.11	2.98	8.31	0.00	0.00	0.00	0.00	0.00	0.51	0.80	0.56	0.26	0.22	-0.09
$ExpScRC^m$	0.04	0.11	2.90	7.71	0.00	0.00	0.00	0.00	0.00	0.49	0.75	0.86	0.46	0.17	-0.22
$ExpScRC^q$	0.04	0.11	2.83	7.10	0.00	0.00	0.00	0.00	0.00	0.49	0.69	0.96	0.73	0.24	-0.38
$ExpScRC^{d-}$	0.06	0.17	2.48	4.55	0.00	0.00	0.00	0.00	0.00	0.64	0.90	0.40	0.20	0.19	-0.10
$ExpScRC^{w-}$	0.06	0.16	2.48	4.60	0.00	0.00	0.00	0.00	0.00	0.61	0.85	0.53	0.25	0.25	-0.09
$ExpScRC^{m-}$	0.06	0.15	2.48	4.58	0.00	0.00	0.00	0.00	0.00	0.59	0.79	0.85	0.46	0.20	-0.22
$ExpScRC^{q-}$	0.06	0.15	2.47	4.51	0.00	0.00	0.00	0.00	0.00	0.57	0.75	0.96	0.74	0.28	-0.38

Notes. The table reports the descriptive statistics of all realized correlation features. The sample consists of 417 stocks listed on NYSE/AMEX/NASDAQ that have ever been included in the S&P 500 index and have full historical quotations over the period from January 2000 to December 2020 with share code 10 or 11, price between \$1 and \$1,000, and daily number of trades greater than or equal to 100. Superscripts d , w , m , and q are abbreviations of daily, weekly, monthly, and quarterly construction intervals or center-of-mass. RC^h and RC^{h-} ($h = d, w, m$) denote the daily, weekly, and monthly realized correlation and negative semicorrelation, respectively. FRC^h ($h = d, w, m$) is the daily, weekly, and monthly factor-driven realized correlation. $ExpRC^h$ and $ExpRC^{h-}$ ($h = d, w, m, q$) are the exponential realized correlation and negative semicorrelation calculated from exponentially weighted moving average of past 500-day realized covariances and negative semicovariances using the corresponding center-of-mass h . $ExpScRC^h$ and $ExpScRC^{h-}$ ($h = d, w, m, q$) are the sector-specific exponential realized correlations and semicorrelations with center-of-mass h . Std, standard deviation.

of predictors used in the forecasting model. This hyperparameter should naturally be tuned based on the prediction error, rather than the training error, to prevent LASSO from overfitting the training sample and performing poorly out-of-sample. Accordingly, we adopt a traditional training-validation-testing scheme for our hyperparameter calibration and model assessment.

To enhance the efficiency of the estimates, rather than fitting the models on a pair-by-pair basis, we instead fit “pooled models” based on the full panel of all the pairwise realized correlations. To help motivate such an approach, Figure 3 plots the 12-month moving averages of the cross-sectional means of the monthly realized correlations for different size quintiles, as measured by the average market capitalization of firm pairs. The dynamic dependencies observed for the five different groups are obviously very similar, highlighting the potential efficiency gains from the use of a pooled panel estimation approach.⁹

More specifically, at the end of year t , we divide the full sample into three parts: a training set consisting of data from year $t - 4$ to year $t - 1$, a validation set consisting of year t data, and a testing set consisting of

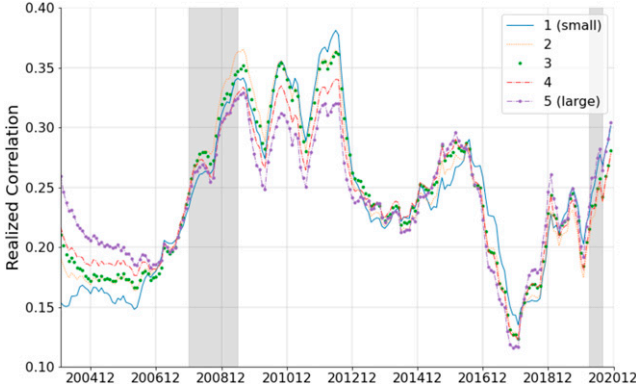
year $t + 1$ data. We then refit the models every year by rolling the training, validation, and testing sets one year forward. Given the $417 \times (417 - 1)/2 = 86,736$ unique stock pairs per month, each 4-year training sample thus includes 4,163,328 observations, allowing us to estimate models with many features. Importantly, our rolling-window scheme also allows the features selected by LASSO to dynamically enter and exit the prediction models based on recent market conditions.

By contrast, our OLS-based prediction models rely on the same set of features throughout. Correspondingly, because the models estimated by OLS do not require validation sets for hyperparameter tuning, we use data from year $t - 4$ to t for their estimation and the data in year $t + 1$ for testing, ensuring identical testing samples for both the OLS- and LASSO-based models.

3.2. Model Fitting and LASSO

All of the forecasting models discussed in the main part of the paper are based on simple linear combinations of the different features—say, $f(x_{ij,t}; \theta) \equiv x'_{ij,t} \theta$, where $x'_{ij,t}$ denotes the feature vector for stock pair

Figure 3. (Color online) Size-sorted Monthly Realized Correlations



Notes. The figure displays 12-month moving averages of the cross-sectional means of monthly realized correlations for the S&P 500 stocks in our sample. The correlations are sorted into size quintiles, with the size for a given stock pair determined by the average market capitalization of the two firms.

(i, j) in month t and θ is the unknown parameter to be estimated. Unlike OLS, LASSO estimates θ through a penalized L_1 loss function. Specifically, to determine the best predictor for the monthly correlation for stock pair (i, j) , LASSO seeks to minimize:

$$\mathcal{L}^{\text{LASSO}}(\theta; \lambda) = \frac{1}{N} \sum_{(ij,t) \in \mathcal{T}} (RC_{ij,t+1} - x'_{ij,t} \theta)^2 + \lambda \sum_{p=1}^P |\theta_p|, \quad (3)$$

where $\mathcal{T}$ denotes a given training sample, N is the number of observations in that training sample, and λ refers to the shrinkage parameter that controls the degrees of penalty on the coefficients. For $\lambda = 0$, the LASSO estimator obviously collapses to standard OLS. However, when $\lambda > 0$, LASSO is capable of setting some of the coefficients to be exactly zero, thereby reducing the parameter estimation error and potentially also making the model easier to interpret.

Meanwhile, to allow for meaningful feature selection, we standardize each predictor to have comparable magnitude, as otherwise a single λ would have vastly different shrinkage effects on different features, making the model impossible to tune. To prevent any look-ahead bias, we therefore normalize each feature by its training-sample mean and standard deviation. Consistent with the rolling-window scheme detailed in the previous section, the mean and standard deviation are calculated once per year. To allow for various levels of sparsity, we consider a wide range of different values of λ , choosing the final preferred model and values of θ and λ from this collection of models.

We turn next to a discussion of the resulting out-of-sample predictive performance obtained by the various OLS- and LASSO-based models.

4. Out-of-Sample Forecast Performance

We begin our assessment of the statistical out-of-sample forecasting performance by evaluating the incremental gains from including additional features beyond the traditional HAR-type features in OLS-based models. We then illustrate the benefits of LASSO over the traditional OLS-based models. Finally, we analyze the sparsity of the LASSO-selected features and their dynamic patterns through time.

4.1. Performance Evaluation Measures

Following common practice in the literature, we focus our statistical assessment of the different models on their out-of-sample R^2 's relative to the HAR model, which, as previously noted, has emerged as the benchmark model for realized volatility-based forecasting. Specifically:

$$R_{\text{OOS}}^{2,EW}(\theta) = 1 - \frac{\sum_{(ij,t) \in \mathcal{T}'} (RC_{ij,t} - \widehat{RC}_{ij,t}^\theta)^2}{\sum_{(ij,t) \in \mathcal{T}'} (RC_{ij,t} - \widehat{RC}_{ij,t}^{\text{HAR}})^2}, \quad (4)$$

where $\widehat{RC}_{ij,t}^\theta$ refers to the forecasts from model θ , $\widehat{RC}_{ij,t}^{\text{HAR}}$ denotes the forecasts based on the traditional HAR features and OLS-based estimation, and $\mathcal{T}'$ defines the specific testing sample.¹⁰ Accordingly, a positive $R_{\text{OOS}}^{2,EW}(\theta)$ indicates that model θ achieves smaller out-of-sample prediction mean squared errors than the benchmark HAR model. To ensure that our evaluation is not primarily driven by small-cap firms, we also calculate a value-weighted version of the R_{OOS}^2 as:

$$R_{\text{OOS}}^{2,VW}(\theta) = 1 - \frac{\sum_{(ij,t) \in \mathcal{T}'} \omega_{ij,t} (RC_{ij,t} - \widehat{RC}_{ij,t}^\theta)^2}{\sum_{(ij,t) \in \mathcal{T}'} \omega_{ij,t} (RC_{ij,t} - \widehat{RC}_{ij,t}^{\text{HAR}})^2}, \quad (5)$$

where $\omega_{ij,t}$ denotes the product of the market capitalizations for stocks i and j normalized to sum to unity.¹¹

To more formally assess the statistical significance of the numerical differences in the out-of-sample R^2 's, we also implement a simple modified Diebold and Mariano (1995) (DM) test for pairwise comparison of two models based on the difference in the out-of-sample squared error losses. Specifically, for stock pair (i, j) in month t , we define the loss differential as $d_{ij,t} = (\hat{e}_{ij,t}^{(1)})^2 - (\hat{e}_{ij,t}^{(2)})^2$, where $\hat{e}_{ij,t}^{(1)}$ and $\hat{e}_{ij,t}^{(2)}$ denote the prediction errors from each of the two models. For each stock k , we then compute the average of the resulting loss differentials across all the stock pairs containing that stock. In other words, we treat each stock as a cross-sectional observation by aggregating the error differentials from all pairs that include the stock. That is:

$$d_k = \frac{1}{N_k T} \sum_{ij,t} d_{ij,t} \mathbf{1}_{\{i=k \text{ or } j=k, i \neq j\}}, \quad (6)$$

where N_k denotes the number of stock pairs containing stock k in each month, and T refers to the total

number of months in the entire testing sample. Our modified DM test statistic is obtained as $DM = \bar{d}/\hat{\sigma}_d$, where $\bar{d}$ and $\hat{\sigma}_d$ denote the cross-sectional mean and standard error of d_k . In parallel with the value-weighted R^2 defined above, we also construct a value-weighted version of this modified DM test statistic by applying the product-based weights to the loss differentials.

4.2. Forecast Performance

Tables 2 and 3 report the resulting performance measures for the three increasingly more complex OLS-based models, together with the LASSO-based predictions outlined in Section 3.2. For ease of reference, the second column summarizes the specific features included in each of the models.

Looking first at Table 2, and the out-of-sample R^2 's, we observe the same ranking for the equal-weighted and the value-weighted out-of-sample R^2 's defined in Equations (4) and (5), respectively, indicating that the results are not driven by the influence of extremely large or small firms. It is noteworthy that all of the models beat the traditional HAR model, albeit in the case of the Semivariance HAR (SHAR) model not by much. Adding the factor-driven realized features FRC 's to the SHAR model results in more noticeable improvements. When we further expand the feature set to include the $ExpRC$ and $ExpScRC$ realized features, the resulting model outperforms all the other OLS-based models by quite wide margins, with the equal-weighted (value-weighted) R^2_{OOS} relative to the HAR equal to 9.82% (7.31%), underscoring the value of incorporating long-memory dynamics and within-sector spillover effects captured by these features.

Having established that the use of our new augmented feature sets can improve the out-of-sample

Table 3. Out-of-sample Predictions: DM t -Statistics

Model	HAR	(1)	(2)	(3)
Panel A: Equal-weighted				
(1) SHAR	11.55			
(2) SHAR-F	29.32	27.58		
(3) SHAR-F-Exp	39.08	39.84	35.24	
(4) LASSO	47.69	48.92	43.41	6.36
Panel B: Value-weighted				
(1) SHAR	4.99			
(2) SHAR-F	13.56	13.41		
(3) SHAR-F-Exp	16.21	16.29	15.51	
(4) LASSO	17.85	17.91	17.41	9.01

Notes. This table and Table 2 report the out-of-sample performance for the OLS-based correlation forecasting models and the LASSO-based correlation forecasting model. The sample consists of 417 stocks listed on NYSE/AMEX/NASDAQ that have ever been included in the S&P 500 index and have full historical quotations over the period from January 2000 to December 2020 with share code 10 or 11, price between \$1 and \$1,000, and daily number of trades greater than or equal to 100. Superscripts d , w , m , and q are abbreviations of daily, weekly, monthly, and quarterly construction intervals. RC^h and RC^{h-} ($h = d, w, m$) denote the daily, weekly, and monthly realized correlation and negative realized correlation, respectively. FRC^h ($h = d, w, m$) is the daily, weekly, and monthly factor-driven realized correlation. $ExpRC^h$ and $ExpRC^{h-}$ ($h = d, w, m, q$) are the exponential realized correlation and negative semicorrelation calculated from exponentially weighted moving average of past 500-day realized covariances and negative semicovariances using the corresponding center-of-mass h . $ExpScRC^h$ and $ExpScRC^{h-}$ ($h = d, w, m, q$) are the sector-specific exponential realized correlations and semicorrelations with center-of-mass h . Our correlation forecasting models include three OLS-based models—SHAR, SHAR-F, and SHAR-F-Exp—and the LASSO model. In this table, Panels A and B report the modified DM t -statistics and value-weighted DM t -statistics for pairwise comparisons among models, respectively.

performance through simple OLS-based fits, the row labeled LASSO demonstrates that additional improvements are available through the use of the LASSO algorithm.¹² Of course, some features may not contribute to

Table 2. Out-of-sample Predictions: R^2_{OOS} Relative to HAR

Model	Feature set	Equal-weighted (%)	Value-weighted (%)
(1) SHAR	3 RC^h + 3 RC^{h-} (# of features = 6)	0.22	0.11
(2) SHAR-F	3 RC^h + 3 RC^{h-} + 3 FRC^h (# of features = 9)	1.71	1.30
(3) SHAR-F-Exp	3 RC^h + 3 RC^{h-} + 3 FRC^h + 4 $ExpRC^h$ + 4 $ExpRC^{h-}$ + 4 $ExpScRC^h$ + 4 $ExpScRC^{h-}$ (# of features = 25)	9.82	7.31
(4) LASSO	All 25 main features	10.16	8.05

Notes. This table and Table 3 report the out-of-sample performance for the OLS-based correlation forecasting models and the LASSO-based correlation forecasting model. The sample consists of 417 stocks listed on NYSE/AMEX/NASDAQ that have ever been included in the S&P 500 index and have full historical quotations over the period from January 2000 to December 2020 with share code 10 or 11, price between \$1 and \$1,000, and daily number of trades greater than or equal to 100. Superscripts d , w , m , and q are abbreviations of daily, weekly, monthly, and quarterly construction intervals. RC^h and RC^{h-} ($h = d, w, m$) denote the daily, weekly, and monthly realized correlation and negative realized correlation, respectively. FRC^h ($h = d, w, m$) is the daily, weekly, and monthly factor-driven realized correlation. $ExpRC^h$ and $ExpRC^{h-}$ ($h = d, w, m, q$) are the exponential realized correlation and negative semicorrelation calculated from exponentially weighted moving average of past 500-day realized covariances and negative semicovariances using the corresponding center-of-mass h . $ExpScRC^h$ and $ExpScRC^{h-}$ ($h = d, w, m, q$) are the sector-specific exponential realized correlations and semicorrelations with center-of-mass h . Our correlation forecasting models include three OLS-based models—SHAR, SHAR-F, and SHAR-F-Exp—and the LASSO model. This table reports R^2_{OOS} and R^2_{OOS} relative to HAR for each model using the entire panel of stock pairs according to Equations (4) and (5).

the forecasts over the entire sample period. The dynamic regularization afforded by the LASSO algorithm naturally helps in the dynamic selection of the most useful features, thus mitigating the risk of overfitting. At the same time, even though LASSO does result in the highest R^2_{OOS} among all of the models, the improvements in the R^2_{OOS} 's compared with the OLS-based SHAR-F-Exp model that always includes all features may appear relatively minor.

Hence, to more formally assess whether these and the other improvements observed in Table 2 are actually statistically significant, Table 3, Panels A and B report the equal- and value-weighted DM t -statistics detailed in Section 4.1 for the pairwise comparisons of the different models. A positive t -statistic in the table indicates that the “row model” outperforms the “column model.” As the results show, all of the t -statistics for the LASSO-based predictions are positive and strongly statistically significant. In other words, LASSO consistently delivers the lowest forecast errors, with statistically significant improvements even over the fully specified SHAR-F-Exp model.

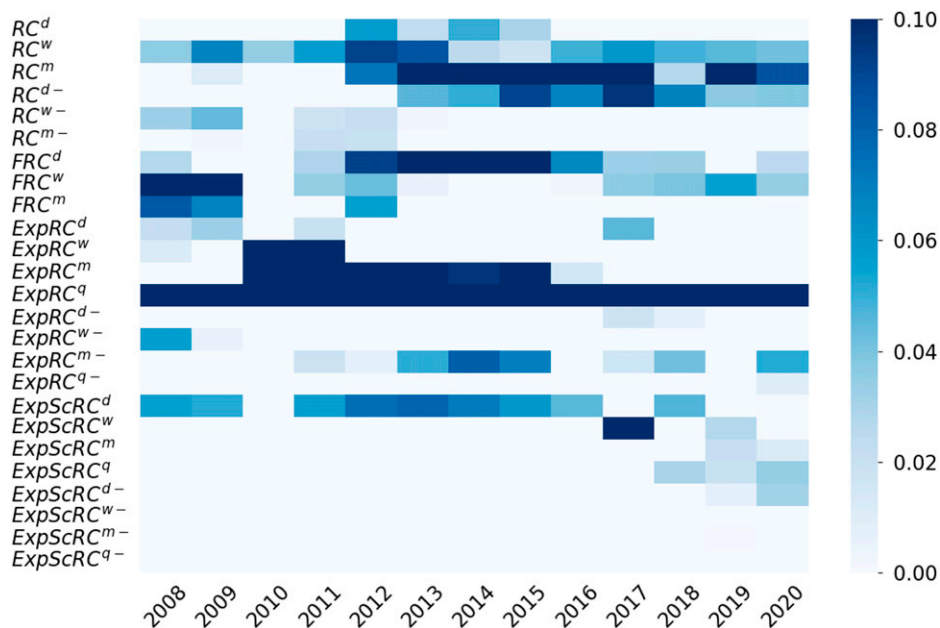
4.3. Feature Selection

The sparsity engendered by the LASSO algorithm also brings with it easier interpretability of the forecasting models, in the form of the dynamic patterns of the selected features and their relative contributions through time.¹³ To illustrate, for each year in the testing sample, we calculate the absolute values of the

estimated LASSO coefficients normalized by the sum of the absolute values across all stock pairs so that the scaled absolute coefficients add up to unity. Figure 4 presents the resulting relative contributions of the 25 realized features across years. The plot confirms that the LASSO models are indeed sparse-encouraging, with 10 features selected per year, on average. Among all realized features, $ExpRC^q$ is the most important predictor, being present in all testing samples and contributing around 50% overall to the predictions as measured by its scaled absolute coefficient. RC^w is also systematically selected, although it contributes less than around 5%, on average. Among the three traditional HAR predictors, the lagged dependent variable RC^m is selected in 10 out of 13 years, with an average importance of around 11%. The second set of frequently selected predictors includes the three short-term signals, FRC^d , FRC^w , and $ExpScRC^d$, each contributing around 4%, meaning that filtering out “noise” through the use of common factor structures and accounting for within-sector spillover effects are both valuable for correlation prediction. Lastly, although $ExpRC^m$ is only selected by LASSO in half of the testing years, its average contribution in those years is more than 15%, substantiating our early conjecture that certain features are only important under certain market conditions.¹⁴

Looking across the columns in Figure 4 also reveals some notable patterns in the relative importance of the different features over time. In particular, fewer

Figure 4. (Color online) Contribution of Selected Features



Notes. The figure displays the scaled absolute coefficients on the 25 main realized features used in the LASSO model. The columns denote each year in the testing sample, and the rows display the 25 features. The color gradients within each column indicate the most influential (dark) to the least influential (light) features, with white indicating features that are not selected.

features are seemingly selected during periods of heightened economic uncertainty. For instance, when trained on data that include the 2007–2009 National Bureau of Economic Research (NBER)-designated recession, LASSO selects only 4 out of the 25 features for the 2010 prediction model. This pattern is intuitive: during times of economic stress, different segments of the market tend to move more closely together due to the increased importance of common economic shocks, thereby reducing the marginal predictive value of some of the individual features. LASSO's data-driven variable selection internalizes this phenomenon, resulting in more parsimonious models during periods of heightened economic uncertainty. Conversely, during more stable economic environments, different dynamic dependencies play a relatively more important role, with a richer set of features meaningfully contributing to the out-of-sample predictions. This directly highlights the strength of the LASSO-based models over their static OLS-based counterparts, the latter which are not engineered to adapt to changing market conditions. Further along these lines, even though several long-term predictors are consistently selected by LASSO for most of our sample, different short-term signals enter and exit the models, in concert with the varying strength of the underlying signals.¹⁵ Nonetheless, many of the selected features remain the same from year to year, underscoring the overall stability of the LASSO-based approach.

5. Applications

The previous section demonstrates that the LASSO-based predictions result in significantly higher out-of-sample R^2 's than all of the OLS-based models, the popular HAR model included. In this section, we further evaluate the economic significance of the predictions by considering four practical applications. In the first application, we construct an augmented pairs trading strategy following Chen et al. (2019) and show that the use of our new forecasting model substantially enhances the strategy's return performance. The second application builds on the finding of Pollet and Wilson (2010) that the average correlation among stocks can predict aggregate market returns. In the third application, we construct a series of long-short portfolios based on popular trading strategies and find that our model significantly improves the consistency between the realized and forecasted risks of the various portfolios. In the last application, we show that covariance matrix forecasts based on our new correlation forecasting models systematically achieve the lowest realized risks for the Global Minimum

Variance portfolio, resulting in substantial utility gains for mean-variance investors.

5.1. Pairs Trading

Pairs trading bets on the price convergence of pairs of stocks, or stocks with strong price comovements. When price divergence occurs, pairs traders take a long position in the underperforming stocks and a short position in the overperforming stocks, hoping to profit from the convergence. Building on Chen et al. (2019), we show how the better correlation forecasts obtained from our new model can help improve the performance of such strategies.

For each stock i in month t , we compute its historical correlation with each of the remaining 416 stocks in our S&P 500 universe, using the time-series average monthly realized correlation over the 12-month window from month $t - 12$ to $t - 1$. We define the top 20 stocks with the highest historical correlation with stock i as its pairs. The historical correlation between stock i and its pair portfolio, denoted $RC_{i,t}^h$, is then calculated as the equal-weighted average of the historical correlations between stock i and each of its pairs.

Let $PRet_{i,t}$ denote the equal-weighted average return of the stocks in stock i 's pair portfolio. Following Chen et al. (2019), we define the return divergence between stock i and its pair portfolio, denoted $RetDiff_{i,t}$, as:

$$RetDiff_{i,t} = \beta_{i,t}(PRet_{i,t} - r_{f,t}) - (Ret_{i,t} - r_{f,t}), \quad (7)$$

where $r_{f,t}$ denotes the risk-free rate, $Ret_{i,t}$ is the return on stock i , and $\beta_{i,t}$ is the estimated slope coefficient from regressing stock i 's returns on its pair portfolio returns using daily data from month $t - 12$ to $t - 1$. The intuition behind the pairs trading strategy is straightforward: if in month t , stock i 's return $Ret_{i,t}$ is higher (lower) than its pair portfolio return $PRet_{i,t}$ after adjusting for risk, the stock is likely overvalued (undervalued) and expected to underperform (outperform) in the following month. To implement this strategy, at the end of each month, we sort stocks into quintile portfolios based on $RetDiff$ and construct a long-short portfolio by buying stocks in the highest $RetDiff$ quintile and selling those in the lowest.

A key implicit assumption behind the pairs trading strategy described above is the persistence of correlations—that is, a pair portfolio constructed based on historical correlations with stock i is expected to continue to closely comove with stock i in month $t + 1$, even if temporary price divergence occurs in month t . However, this assumption may break down, particularly during periods with volatile market conditions. To address this, we enhance the strategy by explicitly incorporating predicted correlations into portfolio construction. Specifically, for stock i in month t , we

compute its predicted correlation with its pair portfolio in month $t + 1$ as the equal-weighted average of its predicted correlations with each of its pair stocks. To keep the presentation focused, we concentrate on predictions from the HAR- and LASSO-based models, though similar results hold for the other forecasting models discussed earlier.

We define the change in correlation as $\Delta RC_{i,t}^\theta = \widehat{RC}_{i,t+1}^\theta - RC_{i,t}^\theta$, where $\widehat{RC}_{i,t+1}^\theta$ denotes the correlation prediction from model θ available at the end of month t . This difference measures the expected shift in comovement between stock i and its pair portfolio and, thus, captures correlation persistence. To implement the improved strategy, we first sort all stocks into quintiles based on $\Delta RC_{i,t}^\theta$ and retain only those in the highest quintile—that is, stocks with the strongest expected increase in correlation with their pair portfolios. Within this subset, we further form a long-short portfolio by buying stocks with high *RetDiff* and selling those with low *RetDiff*.

Table 4, Panels A and B report the annualized equal-weighted and value-weighted average returns of these quintile-sorted portfolios, along with the returns and t -statistics of the spread portfolios.¹⁶ Looking first at the “Unconditional” strategy based on all of the S&P 500 stocks in our sample that does not take into account the persistence of the correlations, we see that it results in rather poorly performing portfolios, with an equal-weighted (value-weighted) spread of 1.15% (−1.20%) per year and a t -statistic of 0.47 (−0.45).¹⁷ By contrast, the long-short portfolios formed by the subset of stocks within the highest ΔRC^θ quintile based on the HAR and LASSO correlation predictions all result in average positive returns. Intuitively, if model θ is good at predicting next month’s correlations, the

return convergence for the resulting pair stocks is likely to be stronger. For stocks within the highest ΔRC quintile based on HAR, the equal-weighted (value-weighted) return spread sorted by *RetDiff* does indeed increase to 3.63% (6.14%) per year, albeit statistically insignificant. Meanwhile, for the stocks within the highest ΔRC quintile based on LASSO, the equal-weighted (value-weighted) return spread increases to an impressive 9.28% (8.83%) per year, with a t -statistic of 2.30 (2.20). This confirms the superior predictive accuracy of the LASSO-based model and highlights the benefit of integrating improved correlation forecasts into pairs trading.

To further underscore the significance of the results, we also conduct a series of predictive Fama-MacBeth cross-sectional regressions, controlling for the four Fama-French-Carhart risk factors.¹⁸ To allow for a direct comparison with the previous single-sorting results, we transform *RetDiff* into five discrete quintile groups. Table 5 reports the resulting estimated regression coefficients and t -statistics under several model specifications. Consistent with our previous portfolio-based analyses, *RetDiff* exhibits the most significant predictive power among stocks with the highest price convergence rate (ΔRC) as predicted by LASSO. For example, take the regressions with control variables reported in the even-numbered columns. The coefficient for *RetDiff* increases from 0.04 (t -statistic 0.97) in column (2) for the full stock sample, to 0.12 (t -statistic 1.80) in column (4) for stocks within the highest ΔRC^{HAR} quintile, to 0.16 (t -statistic 2.31) in column (6) for stocks within the highest ΔRC^{LASSO} quintile. Given that *RetDiff* takes integer values between 1 and 5 representing the five *RetDiff* quintiles, moving up one

Table 4. Performance of Pairs Trading Strategies: Portfolios Sorted by Return Divergence

Sample	1 (Low)	2	3	4	5 (High)	HML
Panel A: Equal-weighted portfolio sorted by return divergence						
Unconditional	6.92%	5.75%	7.09%	6.45%	8.07%	1.15% (0.47)
HAR	9.53%	6.44%	9.25%	7.90%	13.16%	3.63% (0.88)
LASSO	3.55%	5.15%	6.11%	5.76%	12.83%	9.28% (2.30)
Panel B: Value-weighted portfolio sorted by return divergence						
Unconditional	6.05%	4.58%	6.05%	4.76%	4.86%	−1.20% (−0.45)
HAR	6.42%	6.63%	9.02%	7.90%	12.56%	6.14% (1.60)
LASSO	1.88%	5.71%	6.57%	6.39%	10.71%	8.83% (2.20)

Notes. The table and Table 5 report the performance of the pairs trading strategy based on *RetDiff* signals using different stock samples. The full evaluation period is from February 2008 to December 2020. The samples for pairs trading strategy are S&P 500 universe (unconditional sample) and reduced samples based on HAR and LASSO correlation predictions. *RetDiff* signal is the return divergence between stocks and their pair portfolios defined in Equation (7). Panels A and B report the annualized equal-weighted and value-weighted monthly returns of the quintile portfolios sorted by *RetDiff* with different samples. By the end of each month t , we sort stocks on *RetDiff* into five quintile portfolios. We then compute the difference between predicted and historical correlations of a stock’s pair portfolio, $\Delta \text{Corr}^\theta$, based on HAR and LASSO correlation predictions. Lastly, we calculate realized monthly returns in month $t + 1$ for *RetDiff*-sorted portfolios with the unconditional sample and restricted samples of stocks in quintile 5 sorted by $\Delta \text{Corr}^\theta$. The column labeled “HML” reports the difference in returns between Portfolio 5 and Portfolio 1, with t -statistics in parentheses.

Table 5. Performance of Pairs Trading Strategies: Fama-MacBeth Regressions

	Unconditional		HAR		LASSO	
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.50 (1.28)	4.42 (3.78)	0.55 (1.18)	5.79 (2.91)	0.14 (0.32)	6.61 (3.65)
RetDiff	0.03 (0.54)	0.04 (0.97)	0.08 (1.02)	0.12 (1.80)	0.14 (1.85)	0.16 (2.31)
Controls	No	Yes	No	Yes	No	Yes
Adj- R^2	0.59%	12.01%	0.92%	11.73%	1.05%	12.82%
N	64,635	64,635	13,020	13,020	13,020	13,020

Notes. This table and Table 4 report the performance of the pairs trading strategy based on *RetDiff* signals using different stock samples. The full evaluation period is from February 2008 to December 2020. The samples for pairs trading strategy are S&P 500 universe (unconditional sample) and reduced samples based on HAR and LASSO correlation predictions. *RetDiff* signal is the return divergence between stocks and their pair portfolios defined in Equation (7). This table reports the estimated regression coefficients and *t*-statistics (in parentheses) from Fama-MacBeth cross-sectional regressions predicting one-month-ahead stock returns using *RetDiff* defined as discrete values of [1, 2, 3, 4, 5] according to sorted quintiles each month. The control variables include *Beta*, *Size*, *BM*, and *Mom*. Adj- R^2 , adjusted R^2 .

quintile is associated with an increase in annual return of 1.92% (0.16×12).¹⁹

To illustrate how the long-short portfolios based on *RetDiff* perform over time, we compute the cumulative profits of the equal-weighted and value-weighted strategies with an initial investment of $W_1 = \$1$. Specifically:

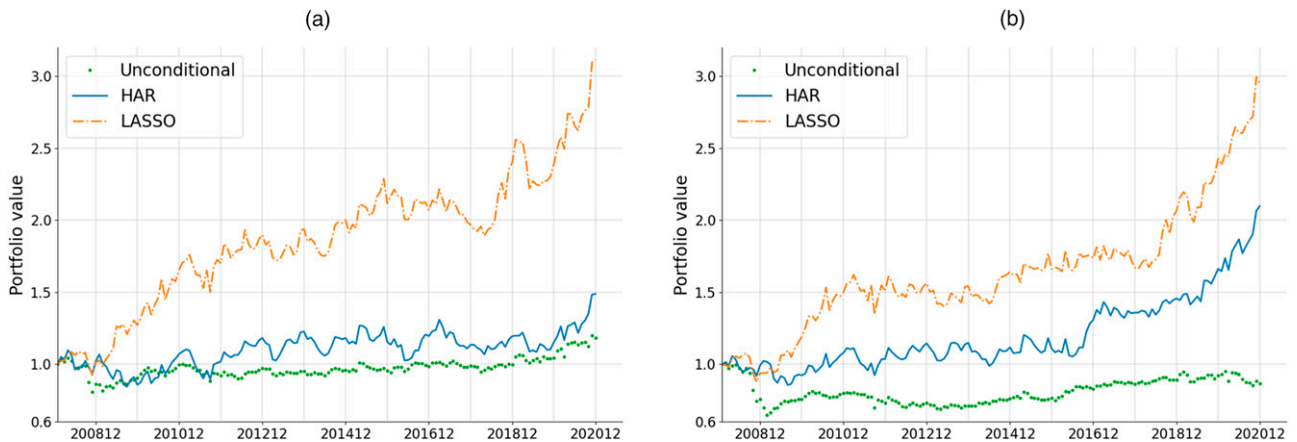
$$W_{t+1} = W_t \times (1 + R_{t+1}^H - R_{t+1}^L + r_{f,t+1}), \quad (8)$$

where R_{t+1}^H and R_{t+1}^L are the monthly returns on the fifth and first quintile portfolios, respectively, for a given strategy. Figure 5, (a) and (b) plot the resulting trajectories of W_{t+1} starting from February 2008. The portfolio value based on stocks within the highest ΔRC^{LASSO} quintile continues to rise throughout the entire sample without major drawdowns. In contrast,

the value of strategies formed by stocks within the highest ΔRC^{HAR} quintile rises only modestly, and those of the unconditional strategies remain almost flat over time. Overall, these results reinforce the benefit of incorporating more accurate correlation forecasts into pairs trading strategies.

5.2. Equity Premium Predictions

Pollet and Wilson (2010) argue that the average correlation among stocks—say, *AvgCorr*—manifests aggregate systematic risks and, therefore, should help predict future aggregate market returns.²⁰ In their empirical analyses, they estimate the expected future average correlation $E_t(AvgCorr_{t+1})$ based on the pairwise correlations computed from lagged daily returns. By the

Figure 5. (Color online) Cumulative Values for Pairs Trading Strategies

Notes. The figure displays the cumulative gains of the pairs trading strategies based on the *RetDiff* signals for different stock samples from February 2008 to December 2020. The samples for the pairs trading strategy include the full S&P 500 universe (Unconditional), the conditional sample based on the correlation predictions from the HAR model (HAR), and the conditional sample based on the correlation predictions from the LASSO-based model (LASSO). The *RetDiff* signal is the return divergence between stocks and their pair portfolios, as defined in Equation (7) in the main text. (a) Equal-weighted. (b) Value-weighted.

same logic, using a more accurate correlation prediction model for $E_t(AvgCorr_{t+1})$ should naturally lead to stronger predictive power for the equity premium.

To test this conjecture, we estimate equal-weighted and value-weighted $E_t(AvgCorr_{t+1})$ from predicted monthly realized correlation for each stock pair as:

$$\begin{aligned} AvgCorr_t^{\theta,EW} &= \sum_{i=1}^N \sum_{j=1, j \neq i}^N \frac{1}{N(N-1)} \widehat{RC}_{ij,t+1}^{\theta} \\ AvgCorr_t^{\theta,VW} &= \sum_{i=1}^N \sum_{j=1, j \neq i}^N \omega_{ij,t} \widehat{RC}_{ij,t+1}^{\theta} \end{aligned} \quad (9)$$

where $\widehat{RC}_{ij,t+1}^{\theta}$ denotes the predicted realized correlation between stocks i and j for month $t+1$, generated by model θ using data available up to month t . The labels *EW* and *VW* refer to equal-weighted and value-weighted average correlations, respectively. N denotes the total number of stocks, and $\omega_{ij,t}$ represents the value-weighting scheme, defined as the product of the market capitalizations of stocks i and j , normalized to sum to one each month.²¹ As before, we focus on predictions from the HAR- and LASSO-based models. For

comparison, we also construct historical equal- and value-weighted average correlations based on realized correlations RC observed in month t .

Table 6 reports the results from predictive regressions for the one-month-ahead excess return on the CRSP value-weighted market index. In addition to the different equal- or value-weighted average correlation measures, we also report results from multiple regressions in which we include the eight commonly used macroeconomic predictors from Welch and Goyal (2008) as controls.²² From the simple regressions in the first three columns of each panel, the $AvgCorr^{H,EW}$ and $AvgCorr^{H,VW}$ constructed from the historical realized correlation are never significant. The $AvgCorr$ measures constructed from the HAR-model predictions exhibit stronger predictive power, albeit still insignificant with t -statistic of 1.47 (1.45) and adjusted R^2 of 0.74% (0.71%) for the equal-weighted (value-weighted) measure in the simple regression without controls. By contrast, the $AvgCorr$ measures constructed from the LASSO-based correlation predictions all provide significant information for predicting future market returns with t -statistics around two and much higher adjusted

Table 6. Equity Premium Predictions

	Panel A: $AvgCorr^{EW}$						Panel B: $AvgCorr^{VW}$					
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.00 (0.05)	−0.02 (−0.94)	−0.02 (−1.41)	0.54 (1.53)	0.49 (1.33)	0.53 (1.52)	0.00 (0.18)	−0.02 (−0.92)	−0.02 (−1.41)	0.55 (1.54)	0.50 (1.35)	0.53 (1.53)
AvgCorr												
RC	0.03 (0.88)			0.03 (0.52)			0.03 (0.75)			0.02 (0.37)		
HAR		0.11 (1.47)			0.08 (0.73)			0.10 (1.45)			0.07 (0.69)	
LASSO			0.13 (2.00)			0.24 (2.40)			0.13 (2.00)			0.24 (2.42)
<i>dp</i>				0.12 (1.65)	0.11 (1.49)	0.13 (1.77)				0.12 (1.66)	0.11 (1.51)	0.13 (1.78)
<i>ep</i>				−0.00 (−0.17)	−0.00 (−0.27)	−0.01 (−0.31)				−0.00 (−0.14)	−0.00 (−0.25)	−0.01 (−0.30)
<i>bm</i>				−0.12 (−0.83)	−0.13 (−0.88)	−0.15 (−0.99)				−0.12 (−0.82)	−0.13 (−0.87)	−0.15 (−0.99)
<i>ntis</i>				0.22 (0.65)	0.20 (0.62)	0.11 (0.33)				0.23 (0.69)	0.21 (0.63)	0.11 (0.32)
<i>tbl</i>				−0.99 (−1.36)	−0.91 (−1.25)	−0.63 (−0.87)				−0.99 (−1.36)	−0.91 (−1.25)	−0.63 (−0.86)
<i>tms</i>				0.11 (0.87)	0.10 (0.83)	0.09 (0.73)				0.12 (0.92)	0.10 (0.85)	0.09 (0.73)
<i>dfy</i>				−2.99 (−1.74)	−2.91 (−1.74)	−4.84 (−2.62)				−2.89 (−1.70)	−2.90 (−1.73)	−4.87 (−2.63)
<i>svar</i>				−0.18 (−0.31)	−0.15 (−0.27)	−0.23 (−0.41)				−0.17 (−0.29)	−0.15 (−0.27)	−0.24 (−0.43)
Adj- R^2	−0.15%	0.74%	1.91%	1.76%	1.94%	5.33%	−0.29%	0.71%	1.92%	1.67%	1.89%	5.38%
<i>N</i>	155	155	155	155	155	155	155	155	155	155	155	155

Notes. The table reports the OLS predictive regression results using different average correlation measures ($AvgCorr^{\theta}$) to forecast future market excess returns. $AvgCorr^{\theta}$ is calculated according to Equation (9) based on a naive model using the lagged correlation as predictors (denoted RC in the table), the HAR model, and the LASSO-based model. The dependent variable is the monthly excess return of the CRSP value-weighted index. The sample spans from February 2008 to December 2020. The control variables include the dividend-price ratio (*dp*), earnings-price ratio (*ep*), book-to-market ratio (*bm*), net equity expansion (*ntis*), treasury-bill rate (*tbl*), term spread (*tms*), default spread (*dfy*), and stock market variance (*svar*), and t -statistics are reported in parentheses. Adj- R^2 , adjusted R^2 .

R^2 's at 1.91% and 1.92%, respectively. Importantly, these same qualitative results for the simple regressions carry over to the multiple regressions with controls. In particular, only the $AvgCorr^{LASSO}$ measures positively and significantly predict next month's market excess return, with t -statistics of 2.40 and 2.42 for the equal- and value-weighted measures, respectively. Given that the average cross-sectional standard deviation of $AvgCorr^{LASSO,EW}$ ($AvgCorr^{LASSO,VW}$) is 0.06, this means that a one-standard-deviation increase in the average correlation predicts a nontrivial rise in the market excess return of $0.06 \times 0.24 \times 12 = 17.2\%$ per annum. Taken together, these findings again underscore the economic value of more accurate correlation forecasts produced by the LASSO-based RC model.

Complementing the previous results, we also construct out-of-sample equity premium forecasts based on an expanding-window framework. Specifically, we estimate the forecasting models based solely on historical or predicted realized correlations, beginning with the 10-year window from 2008 to 2017. We then use these initial estimates to generate the market return forecasts for 2018, subsequently updating the estimation window by one year at a time to generate the forecasts for 2019 and 2020. As further detailed in Table A.5 of the Online Appendix, assessing the forecast performance in terms of the resulting out-of-sample R^2 's relative to the prevailing mean benchmark (as in, e.g., Dong et al. 2022), the predictive signals based on the historical realized correlations (RCs) again underperform those derived from the HAR- and LASSO-based models. Moreover, Diebold-Mariano test statistics further indicate that the improvements afforded by the LASSO-based forecasts are also statistically significant at the usual 5% level, whereas the HAR-based forecasts are not.

5.3. Risk Targeting

Having access to more accurate correlation predictions allows portfolio managers to better assess the risk of their strategies while maximizing returns to investors. Specifically, using the decomposition in Equation (2), one may readily calculate one-month-ahead forecasts for the covariance matrix of monthly returns as:

$$\widehat{RCov}_t^\theta = \sqrt{\widehat{RV}_t} \cdot \widehat{RC}_t^\theta \cdot \sqrt{\widehat{RV}_t}, \quad (10)$$

where $\widehat{RV}_t$ denotes the diagonal matrix of monthly predicted realized variances, and $\widehat{RC}_t^\theta$ denotes the monthly predicted correlation matrix from model θ . To isolate the role of correlation forecasts, we intentionally use the popular HAR model to predict the monthly volatilities for all specifications, regardless of the model used for forecasting the correlations. Although more sophisticated volatility forecasting methods, including machine learning approaches (e.g., Li and Tang 2025), are available, the HAR model remains highly effective

and is widely regarded as the benchmark in the realized volatility forecasting literature. Accordingly, we do not attempt to optimize the volatility forecasts $\widehat{RV}_t$, but instead focus our comparison on the performance of different correlation forecasts $\widehat{RC}_t^\theta$, with particular attention to the HAR- and LASSO-based models.

Given the forecasted covariance matrix $\widehat{RCov}_t^\theta$ for model θ , the expected risk of a portfolio with portfolio weights ω_t under model θ is readily calculated as $\omega_t' \widehat{RCov}_t^\theta \omega_t$. Once the actual portfolio return for month t is realized, the manager may assess the ex post risk of the portfolio and compare it to the forecasted risk using the risk-targeting ratio: $\omega_t' \widehat{RCov}_t^\theta \omega_t / \omega_t' RCov_t \omega_t$. Averaging this ratio over the entire testing period yields a summary measure of the model's risk-targeting accuracy:

$$AvgRatio^\theta = \frac{1}{T} \sum_{t=1}^T \frac{\omega_t' \widehat{RCov}_t^\theta \omega_t}{\omega_t' RCov_t \omega_t}. \quad (11)$$

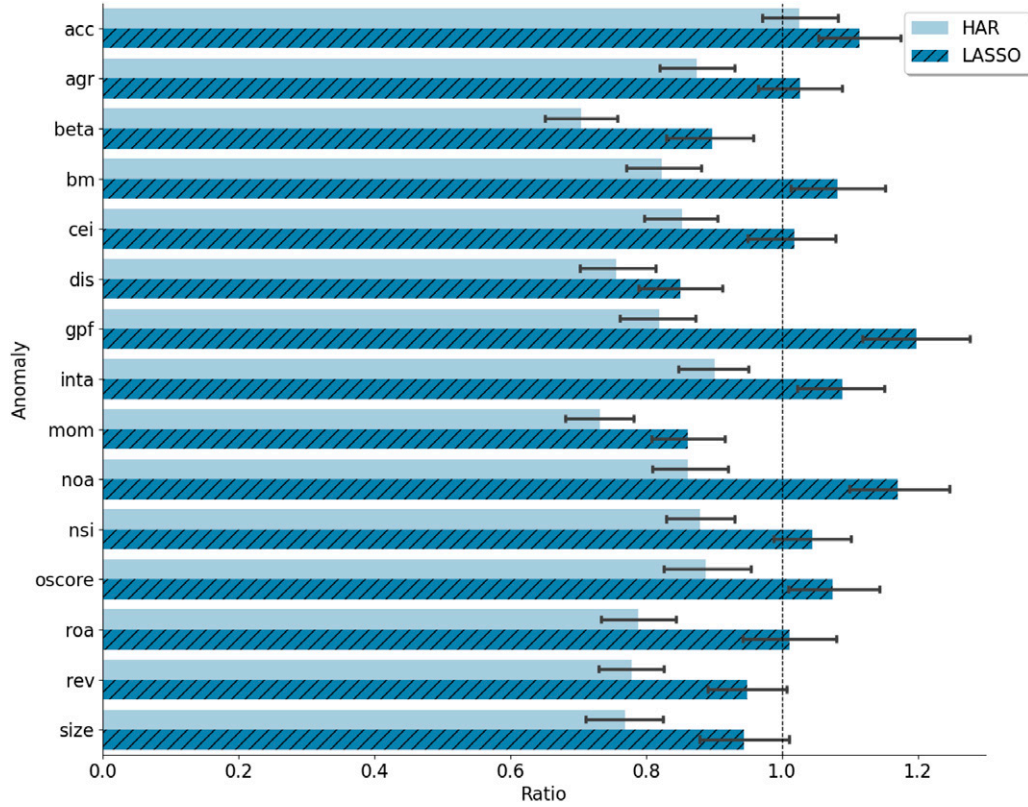
Because the portfolio weights are fixed in advance and all models by design rely on the identical forecasts for the realized variances, any differences in the risk-targeting ratio stem solely from the accuracy of the correlation forecasts. Thus, the closer the average risk-targeting ratio is to unity, the more accurate the correlation predictions are.

We proceed to compare the $AvgRatio^\theta$ ratios defined in (11) for the HAR- and LASSO-based correlation forecasting models. We consider 15 different long-short strategies formed by the anomalies used in our feature construction, as further detailed in the appendix. As the results in Figure 6 clearly show, the risk-targeting ratios for the LASSO-based forecasts are generally much closer to unity than the ones for the HAR-based forecasts.²³ Specifically, whereas the overall mean of the average risk-targeting ratios across all of the strategies equals 1.02 for LASSO, it equals 0.83 for HAR, indicative of systematic underestimation of the true risks. Further along these lines, the target of unity is included in the bootstrapped 95% confidence intervals for six strategies for the LASSO-based forecasts, whereas for only one strategy is unity included in the 95% confidence intervals for the HAR-based forecasts.²⁴ These results thus yet again underscore the superiority of the LASSO-based forecasts in the context of portfolio risk prediction and targeting.

5.4. Global Minimum Variance Portfolio

In our final application, we consider estimates of the Global Minimum Variance portfolio. The GMV is often used for evaluating covariance matrix forecasts, as the portfolio weights only depend on the covariance matrix and not on the expected return, thus

Figure 6. (Color online) Risk-targeting Ratios of Long-short Strategies



Notes. The figure displays the ratios of the forecasted portfolio risk over the realized portfolio risk from the HAR- and LASSO-based models for the 15 main long-short trading strategies discussed in the main text. The ratios are averaged over all testing samples according to Equation (11). The black lines correspond to 95% bootstrapped confidence intervals.

allowing for a “clean” comparison of different risk models.²⁵

Specifically, consider a risk-averse investor who allocates her wealth across N stocks, aiming to minimize the total portfolio risk based on the covariance matrix forecast $\widehat{RCov}_t^\theta$ from some forecasting model θ . By standard mean-variance optimization, her optimal portfolio weight vector is given by:

$$\omega_t^\theta = \frac{(\widehat{RCov}_t^\theta)^{-1}}{\mathbf{1}'(\widehat{RCov}_t^\theta)^{-1}\mathbf{1}}, \quad (12)$$

where $\mathbf{1}$ denotes an $N \times 1$ vector of ones. Notably, this optimal solution requires the estimated covariance matrix to be positive definite. In cases where the forecasted correlation matrix is not positive definite, we apply a simple convexity correction procedure, as detailed in the Online Appendix.²⁶ Calculating these optimal weights for each month t in the testing samples, we construct the resulting monthly portfolio returns $\omega_t^{\theta'} r_t$, realized portfolio risks $\sqrt{\omega_t^{\theta'} RCov_t \omega_t^\theta}$, and portfolio Sharpe ratios $(\omega_t^{\theta'} r_t - r_{f,t}) / \sqrt{\omega_t^{\theta'} RCov_t \omega_t^\theta}$. To keep the presentation manageable, we again focus on the

HAR- and LASSO-based correlation forecasting models, together with the traditional HAR model for forecasting the volatilities, in line with the analysis in Section 5.3.

To further quantify the economic gains, we adopt the utility-based framework of Fleming et al. (2003), as extended by Bollerslev et al. (2018a). Specifically, we assume that the investor’s realized utility from the month- t return under model θ is given by:

$$U(r_t^\theta, \gamma) = (1 + r_t^\theta) - \frac{\gamma}{2(1 + \gamma)} (1 + r_t^\theta)^2, \quad (13)$$

where γ refers to her level of risk aversion. Accordingly, the economic value of switching from forecasting model θ_1 to model θ_2 may naturally be measured by solving for Δ_γ in:

$$\sum_{t=1}^T U(r_t^{\theta_1}, \gamma) = \sum_{t=1}^T U(r_t^{\theta_2} - \Delta_\gamma, \gamma), \quad (14)$$

with Δ_γ interpretable as the return the investor would be willing to give up for using the better correlation forecasting model.

The summary table included in Figure 7 presents the resulting average realized portfolio returns, standard deviations, and Sharpe ratios, together with the

estimates of Δ_γ .²⁷ Although expected returns are not used in the GMV optimization, the average realized monthly return for the LASSO-based portfolios turns out to be slightly higher than for the HAR portfolios. This, of course, is merely by “luck.” Importantly, however, the LASSO-based portfolios also result in lower average risk than the HAR portfolios, with an average portfolio standard deviation of 9.96% versus 10.51%. To more directly illustrate this, Figure 7 provides a scatter plot of the monthly portfolio standard deviations (annualized) for the HAR- and LASSO-based GMV portfolios. As the figure shows, most of the data points are above the 45-degree line, indicating that the LASSO-based GMV portfolios typically result in lower realized risks than the HAR-based GMV portfolios.²⁸ The lower risk for the LASSO-based portfolios, together with the slightly higher average returns, naturally result in a higher average Sharpe ratio of 1.65, compared with 1.23 for the HAR-based portfolios. This also translates into nontrivial economic gains. In particular, from the last three columns in the table, an investor with risk aversion $\gamma = 2$ would be willing to give up 0.77% return per annum for access to our LASSO-based forecasts versus relying on the HAR-based forecasts. This economic gain further increases to 0.98% (1.37%) for a more risk-averse investor with $\gamma = 5$ ($\gamma = 10$).

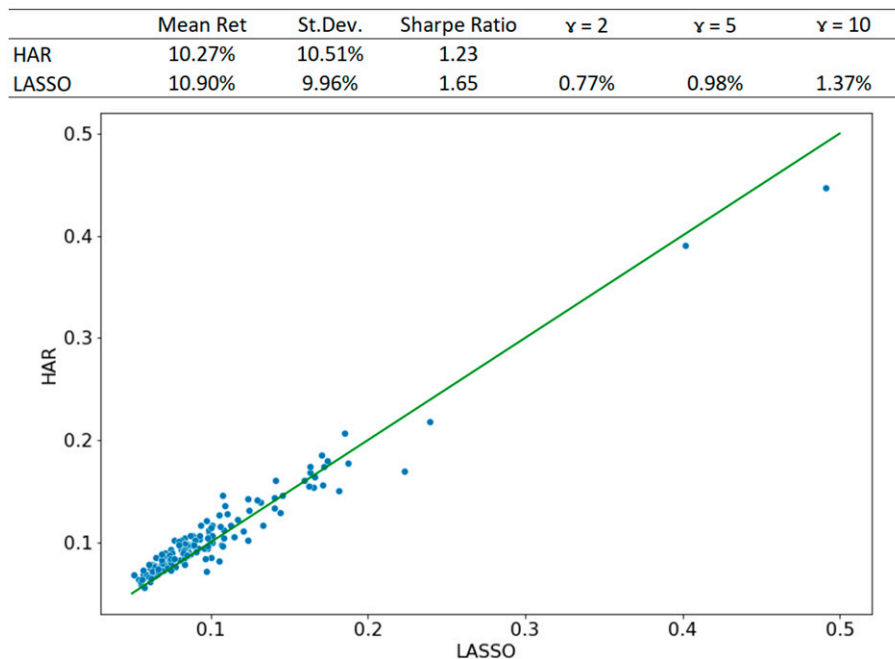
Motivated by a prototypical market-neutral strategy in which managers want to neutralize market risk, we

also consider a beta-neutral GMV analysis following Cosemans et al. (2016). In this setup, we augment the traditional GMV optimization problem with the additional constraint that the portfolio’s beta equals zero. That is:

$$\frac{\omega'_t \widehat{RCov}_t^\theta m_t}{m'_t \widehat{RCov}_t^\theta m_t} = 0, \quad (15)$$

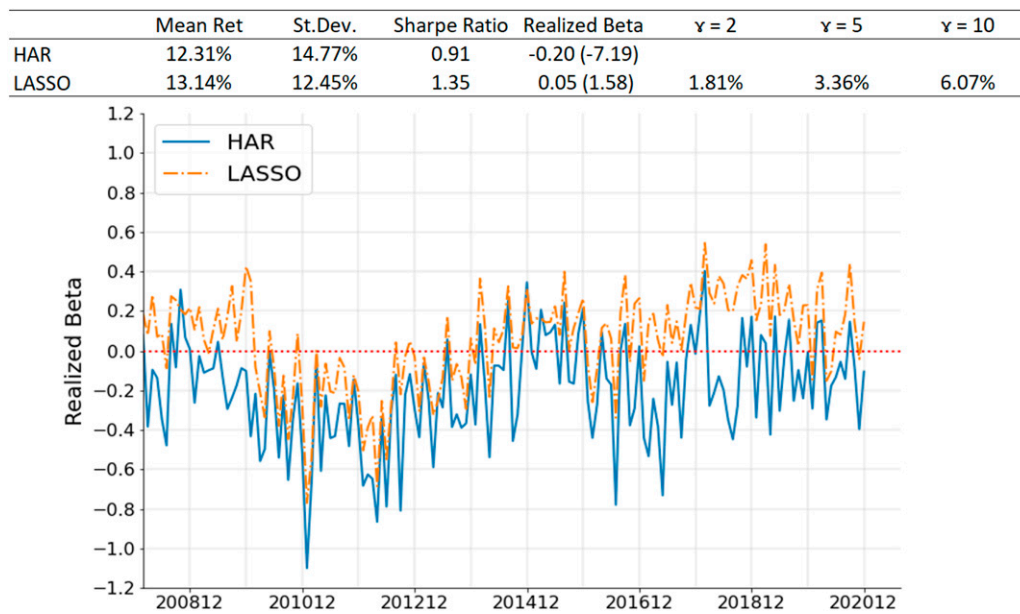
where m_t denotes the $N \times 1$ vector of firm market capitalizations normalized to sum to unity. The returns, risks, and Sharpe ratios of the resulting beta-neutral GMV portfolios reported in the table, included in Figure 8, mirror the patterns for the original GMV portfolios. In particular, the LASSO-based portfolios again attain the lowest average risk and highest average Sharpe ratio. Importantly, the average of the actually realized monthly betas from the LASSO-based predictions, calculated as $\omega'_t \widehat{RCov}_t m_t / m'_t \widehat{RCov}_t m_t$, equal 0.05 with a robust t -statistic of 1.58, suggesting that the portfolios are indeed market neutral. By contrast, the average realized beta for the HAR-based predictions equals -0.20 with a robust t -statistic of -7.19 , indicating significant negative exposure to market risk. To further illustrate these differences, Figure 8 displays time series plots of the monthly realized betas. As the figure shows, the realized betas for LASSO are seemingly symmetric around zero, whereas the betas for HAR

Figure 7. (Color online) GMV Portfolio Comparisons



Notes. The figure compares the performance of GMV portfolios for the HAR- and LASSO-based forecasting models. The upper panel reports the full sample average returns (Mean ret), standard deviations (St.Dev.), and Sharpe ratios for the GMV portfolios, together with the utility gains (in annualized returns) of switching from HAR- to LASSO-based forecasts. The scatter plot in the lower panel displays the (annualized) monthly standard deviations from the two different models.

Figure 8. (Color online) Beta-neutral GMV Portfolio Comparisons



Notes. The figure compares the performance of beta-neutral GMV portfolios for the HAR- and LASSO-based forecasting models. The top panel reports the full sample average returns (Mean ret), standard deviations (St.Dev.), Sharpe ratios, and realized betas for the portfolios, together with the utility gains (in annualized returns) of switching from the HAR- to the LASSO-based forecasts. The time-series plot displays the monthly realized betas for the beta-neutral GMV portfolios from the two different forecasting models.

appear systematically below zero. Interestingly, the imposition of the beta-neutral constraint also substantially increases the utility gains to 1.81% (6.07%) per annum for an investor with risk aversion $\gamma = 2$ ($\gamma = 10$). In other words, not only do the LASSO-based correlation forecasts allow a portfolio manager to better attain her risk target, but they also enable more effective control over market risk exposure.

6. Robustness

6.1. Subsample Analysis

The empirical analyses discussed above relied on the full testing sample for comparing and contrasting the forecasting models. To illustrate that the superior performance remains intact across different subsamples, we divide the full testing sample into three subperiods: 2008–2011, 2012–2015, and 2016–2020.

The resulting out-of-sample performance for each of the three periods is summarized in Table 7. Looking first at Panel A and the equal-weighted R^2_{OOS} 's, we see that during the first subperiod between 2008 and 2011, which covers the financial crisis and its aftermath, LASSO again achieves the highest relative R^2_{OOS} of 7.87%, whereas the OLS-based model with all 25 main features included has the second highest relative R^2_{OOS} of 6.95%. In other words, the sparsity introduced by LASSO enhances predictive accuracy in volatile markets by concentrating on the most informative features. For the second subperiod between 2012 and 2015, LASSO remains the best model with a relative

R^2_{OOS} of 10.70%. For the most recent period spanning 2016 to 2020, although the R^2_{OOS} for LASSO is slightly below the OLS-based model that involves all the features, it clearly outperforms the HAR-based forecasts with an impressive relative R^2_{OOS} of 11.51%. The value-weighted relative R^2_{OOS} 's reported in Panel B evidence a similar pattern. The LASSO-based forecasts generally dominate the OLS-based forecasts, except for the very last subperiod, where it performs on par with the most general OLS-based model.

Motivated by the larger relative gains observed for LASSO during earlier periods of heightened market volatility, we also perform a more detailed analysis pertaining to the recent COVID-19 period. Although the data for the COVID-19 period never enter our training and validation samples, LASSO is purposely designed to avoid overfitting the “noise,” thereby hopefully attaining more robust predictions during that period as well. To corroborate this conjecture, Figure 9 presents a scatter plot of the average realized and predicted correlations for each of the stocks in our sample during the peak six-month COVID-19 period from March 2020, when the World Health Organization declared COVID-19 a pandemic, through August 2020, a half-year later. The reported average correlations for each of the stocks are simply calculated as the mean of all the pairwise correlations that contained a particular stock. As the figure shows, most of the LASSO-based correlation predictions are indeed closer to the 45-degree line than the HAR-based predictions, reaffirming that

Table 7. Out-of-sample Predictions for Different Subsamples

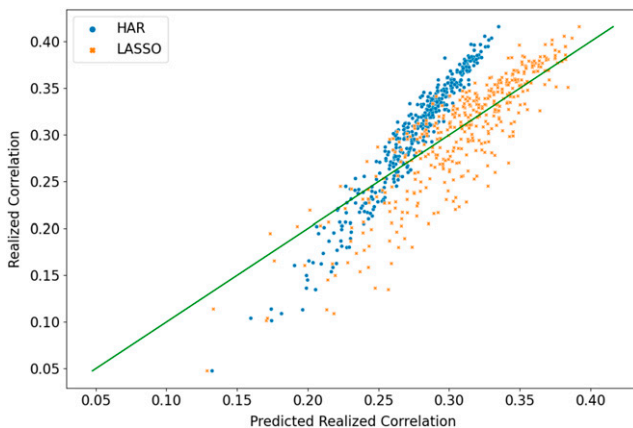
		R^2_{OOS} relative to HAR (%)		
		2008–2011	2012–2015	2016–2020
Model	Feature set			
Panel A: Equal-weighted				
(1) SHAR	3 RC^h + 3 RC^{h-} (# of features = 6)	0.12	0.33	0.23
(2) SHAR-F	3 RC^h + 3 RC^{h-} + 3 FRC^h (# of features = 9)	2.34	0.64	1.97
(3) SHAR-F-Exp	3 RC^h + 3 RC^{h-} + 3 FRC^h + 4 $ExpRC^h$ + 4 $ExpRC^{h-}$ + 4 $ExpScRC^h$ + 4 $ExpScRC^{h-}$ (# of features = 25)	6.95	9.95	11.89
(4) LASSO	All 25 main features	7.87	10.70	11.51
Panel B: Value-weighted				
(1) SHAR	3 RC^h + 3 RC^{h-} (# of features = 6)	0.08	0.25	0.05
(2) SHAR-F	3 RC^h + 3 RC^{h-} + 3 FRC^h (# of features = 9)	2.24	0.04	1.44
(3) SHAR-F-Exp	3 RC^h + 3 RC^{h-} + 3 FRC^h + 4 $ExpRC^h$ + 4 $ExpRC^{h-}$ + 4 $ExpScRC^h$ + 4 $ExpScRC^{h-}$ (# of features = 25)	3.66	9.40	8.47
(4) LASSO	All 25 main features	5.76	10.10	8.31

Notes. The table reports the out-of-sample performance for OLS- and LASSO-based correlation forecasting models for the three subsample periods: 2008–2011, 2012–2015, and 2016–2020. The sample consists of 417 stocks listed on NYSE/AMEX/NASDAQ that have ever been included in the S&P 500 index and have full historical quotations over the period from January 2000 to December 2020 with share code 10 or 11, price between \$1 and \$1,000, and daily number of trades greater than or equal to 100. Superscripts d , w , m , and q are abbreviations of daily, weekly, monthly, and quarterly construction intervals. RC^h and RC^{h-} ($h = d, w, m$) denote the daily, weekly, and monthly realized correlation and negative realized correlation, respectively. FRC^h ($h = d, w, m$) is the daily, weekly, and monthly factor-driven realized correlation. $ExpRC^h$ and $ExpRC^{h-}$ ($h = d, w, m, q$) are the exponential realized correlation and negative semicorrelation calculated from exponentially weighted moving average of past 500-day realized covariances and negative semicovariances using the corresponding center-of-mass h . $ExpScRC^h$ and $ExpScRC^{h-}$ ($h = d, w, m, q$) are the sector-specific exponential realized correlations and semicorrelations with center-of-mass h . Our correlation forecasting models include three OLS-based models—SHAR, SHAR-F, and SHAR-F-Exp—and the LASSO model. Panels A and B report R^2_{OOS} and $R^2_{OOS}^{VW}$ relative to the HAR model, respectively.

LASSO consistently delivers more accurate forecasts, including during extreme market conditions.

6.2. Alternative Features and Machine Learning Techniques

The 25 features underlying our main results all involve various realized correlation-type measures. As discussed

Figure 9. (Color online) Out-of-sample Predictions During the Peak of COVID-19

Notes. The figure shows for each stock in our sample the average realized correlations (y-axis) against the average predicted correlations (x-axis) from the HAR model and the LASSO-based model between March 2020 and August 2020. The average correlation for a stock is calculated as the mean of all the pairwise correlations that contain the stock.

above, our use of these measures as our main features is naturally motivated by earlier findings in the financial econometrics literature. Meanwhile, a number of different economically motivated measures have also previously been proposed in the finance literature for capturing firm connections. Concretely, we consider the distance between two firms' headquarters as measured by their zip code distance (Parsons et al. 2020), economic activity as measured by text-based network industry classifications (Hoberg and Phillips 2010, 2016), industry supply chain dependence (Menzly and Ozbas 2010), common analyst coverage (Israelson 2016), common active mutual fund ownership (Antón and Polk 2014), and common passive mutual fund ownership (Appel et al. 2016).²⁹ Table A.6 in the Online Appendix provides more detailed definitions and data sources.

To investigate whether any of these six additional economically motivated firm-linkage variables may be used to further improve the out-of-sample correlation forecasts, we begin by summarizing each variable using a simple dummy based on the median split, the only exception being the dummy for text-based industry classification (TNIC3), which already takes a value of one if two firms are classified into the same industry and zero otherwise. Armed with these dummies, we then consider two new augmented feature sets, with the first set consisting of our original 25 features plus the 6 dummies, and the second set consisting of the 25 main features plus 150 interaction terms

formed by interacting each of the original features with the six dummies. This setup allows us to capture both first-order direct effects and second-order interaction effects.

In addition to these alternative feature sets, we also consider the use of alternative machine learning algorithms for the construction of the prediction models. In particular, although our main results suggest that LASSO is effective in reducing the number of parameters in the forecasting models through the use of an L_1 -norm penalty, it assumes a linear relation between the forecasts and the features. It is possible that this linearity may be too restrictive, and relaxing it could lead to better predictive performance. Hence, to investigate this, we implement the following four additional machine learning algorithms: Ridge Regression (Ridge), Elastic Net (ENet), Principal Component Regression (PCR), and Feed-forward Neural Networks (FNN).³⁰ The first three of these allow us to explore different regions of the parameter space in linear models, while the last allows us to more thoroughly explore possible nonlinear relations and interactions.³¹

Table 8 reports the R^2_{OOS} relative to the simple HAR forecasts obtained by different combinations of the feature sets and machine learning algorithms. We purposely maintain the same feature standardization and training scheme underlying the results reported in Tables 2 and 3 to allow for a direct comparison with the earlier results.³² In line with the earlier results, the equal and value-weighted R^2_{OOS} 's reported in Panels A and B, respectively, are fairly similar, suggesting that our findings are not dominated by many small firms and/or a few large firms. Looking across the rows, for the LASSO-, Ridge-, and ENet-based

predictions, we observe only minor increases in the R^2_{OOS} 's by adding the new firm-linkage dummies, whether directly or interactively. As a case in point, for the LASSO-based predictions, the equal-weighted relative R^2_{OOS} increases from 10.16% for the model based on our original 25 features, to just 10.24% and 10.35% for the models based on the new augment feature sets, indirectly suggesting that the relevant information about firm linkages for correlation prediction are already embedded in the time-series based realized features. Further corroborating this conjecture, for both the PCR- and FNN-based predictions, the inclusion of the additional firm-linkage measures actually results in lower R^2_{OOS} 's compared with the prediction models based on our original 25 features only.

Looking across the columns, as expected, the performance of ENet is roughly comparable to that of LASSO, whereas the performance of Ridge appears slightly inferior. In contrast to LASSO, which is based on the sparsity-encouraging L_1 -penalty, Ridge relies on an L_2 -penalty, and thus never enjoys the benefits of true dimension reduction. By comparison, PCR does benefit from dimension reduction, provided that the underlying feature space is well-captured by the selected principal components. However, when this is not the case, PCR tends to underperform LASSO, as evidenced by the R^2_{OOS} 's for the third and largest feature set. Interestingly, the more complicated-to-implement FNN-based models also typically underperform LASSO.

In sum, the results in Table 8 further underscore the superior performance of our simple-to-construct LASSO-based correlation forecasting models, both in

Table 8. Out-of-sample Predictions for Alternative Fitting Procedures

Feature set	R^2_{OOS} relative to HAR (%)				
	LASSO	Ridge	ENet	PCR	FNN
Panel A: Equal-weighted					
All 25 main features	10.16	9.83	10.14	10.44	9.63
All 25 main features + 6 dummies (# of features = 31)	10.24	9.96	10.19	9.61	9.81
All 25 main features + 150 feature × dummy combinations (# of features = 175)	10.35	9.95	10.31	8.76	9.56
Panel B: Value-weighted					
All 25 main features	8.05	7.31	8.07	8.31	6.76
All 25 main features + 6 dummies (# of features = 31)	8.05	7.38	8.09	7.66	6.97
All 25 main features + 150 feature × dummy combinations (# of features = 175)	8.20	7.54	8.23	7.68	6.64

Notes. The table reports the out-of-sample performance of different machine learning correlation forecasting models using alternative feature sets. The sample consists of 417 stocks listed on NYSE/AMEX/NASDAQ that have ever been included in the S&P 500 index and have full historical quotations over the period from January 2000 to December 2020 with share code 10 or 11, price between \$1 and \$1,000, and the daily number of trades greater than or equal to 100. The features include 25 predictors used in the main analysis, six dummies based on firm-linkage variables, and 150 feature-dummy interactive terms. Our correlation forecasting models include LASSO, Ridge Regression (Ridge), Elastic Net (ENet), Principal Component Regression (PCR), and two-hidden-layer Feed-Forward Neural Network (FNN). Panels A and B report R^2_{OOS} and R^2_{OOS} relative to HAR, as defined in Equations (4) and (5), respectively, for each model using different sets of features.

Table 9. Out-of-sample Performance for Extended Sample

Model		Panel A: Refitting		Panel B: Transfer learning	
		R^2_{OOS} relative to HAR (%)		R^2_{OOS} relative to HAR (%)	
		Equal-weighted	Value-weighted	Equal-weighted	Value-weighted
(1)	SHAR	0.25	0.15	0.21	0.09
(2)	SHAR-F	1.55	1.25	1.53	1.14
(3)	SHAR-F-EXP	9.61	7.52	9.65	7.26
(4)	LASSO	9.95	8.13	9.90	7.82

Notes. This table reports the out-of-sample forecast performance for OLS-based and LASSO-based correlation forecasting models. The sample consists of 946 stocks that were ever included in the S&P 500 index over the period from January 2000 to December 2020. Panel A reports the equal-weighted and value-weighted R^2_{OOS} values relative to the HAR models, where all models are re-estimated using the expanded sample. Panel B reports the equal-weighted and value-weighted R^2_{OOS} values relative to the HAR models when all models are directly transferred from the original sample.

terms of the use of alternative firm-linkage measures and more complex machine learning algorithms. At the same time, it is certainly possible that the use of forecast combination techniques (see Timmermann 2006 for a comprehensive discussion) that combine the forecasts from entirely different procedures with the forecasts from the LASSO-based models developed here could result in further improvements and even better economic decisions.

6.3. Extended Unbalanced Sample

Our main analysis focuses on the sample of stocks that were ever constituents of the S&P 500 index and have full historical data between January 2000 and December 2020. This focus on a “balanced panel” was motivated by simplicity and ease of implementation in some of our applications. To alleviate concerns about potential survivorship bias, we expand the stock universe to include all historical constituents of the S&P 500 index from 2003 to 2020, increasing the number of stocks from 417 to 946, and construct the same set of 25 realized features used in the original analysis.

Using this broader data set, we first evaluate the robustness of our findings by re-estimating the same models. Table 9, Panel A reports the R^2_{OOS} relative to the HAR forecasts, where all models, including the benchmark models, are refitted for the expanded sample. In line with our earlier findings, the LASSO-based models continue to outperform their OLS-based counterparts, suggesting that none of our main findings are materially affected by sample selection. Importantly, as shown in Figure A.5 in the Online Appendix, the features selected by LASSO also exhibit a strong overlap with those selected in the original balanced sample shown in Figure 4. This again is consistent with our argument in favor of pooled models that exploit the shared structure in correlation dynamics across stocks.

Motivated by the observed commonality in feature selection and correlation behavior, we also conduct a “transfer learning analysis,” in which we train the models on the original 417-stock universe and then

apply said models directly to all the stocks in the expanded sample. Table 9, Panel B reports the resulting R^2_{OOS} relative to the “transferred” HAR-based forecasts. Once again, the LASSO-based models demonstrate the strongest predictive power, supporting the idea that correlation forecasting models trained on a representative stock universe generalize well to broader sets of stocks.

In sum, these additional analyses confirm that our main results are not driven by survivorship bias or the selection of “winning” stocks. Rather, the identified predictive patterns and model performance persist, even when the universe is broadened to include all “losing” historical constituents.

7. Conclusion

Exploiting techniques and ideas from the recent financial forecasting literature, we design new features and LASSO-based models explicitly geared toward correlation forecasting. Implementing the new models with a large panel of S&P 500 stocks, we document statistically significant improvements in out-of-sample forecast accuracy, along with nontrivial economic gains from utilizing the forecasts from the new models in a wide range of practical applications. We further attribute the success of our preferred models to their ability to adapt to changing market conditions by allowing the many different features to dynamically enter and exit the predictions. Although it is possible that even better forecasting models may be obtained through the use of more complex fitting algorithms, such as tree-based models, alongside richer feature sets, our deliberate focus on linear, economically motivated models is aimed at providing greater transparency and easier interpretability of the results. Indeed, the techniques and insights developed here could easily be extended to construct forecasting models for other commonly used risk measures, including measures of precision and factor risk exposures.

All the results in the paper are based on realized covariance matrices computed using cross-products

of 15-minute high-frequency returns. This relatively coarse sampling frequency is deliberately chosen to reduce measurement biases stemming from market microstructure noise. It is possible that the use of more sophisticated realized estimators (e.g., Aït-Sahalia et al. 2010, Bollerslev et al. 2019) may lead to even better forecasting models. The documented heterogeneity in predictive performance across firm characteristics (e.g., greater gains for high-momentum stocks) also suggests that additional improvements may be available from the use of characteristic-based dummies or more advanced clustering methods.

Another promising avenue for future research entails the use of realized correlation forecasts to study the correlation risk premium. Driessen et al. (2009) compare the variance risk premium (VRP) of a broadly defined market index and its constituents and conclude that the significant VRP generally observed for the market is primarily driven by correlation risk because the VRPs for most individual stocks are not significantly different from zero. Other studies rely on correlation innovation (e.g., Krishnan et al. 2009), or the difference between option-implied and realized correlations (e.g., Driessen et al. 2013, Mueller et al. 2017, Bondarenko and Bernard 2024) to further elucidate the origins of the correlation risk premium. Our new framework for more accurate realized correlation forecasting also holds the promise of shedding new light on the economic forces that drive changes in correlation risk. Along these lines, other potentially interesting empirical applications of the new methods include the prediction of factor betas (e.g., Buss and Vilkov 2012) and the evaluation market exposure and timing ability (e.g., Buraschi et al. 2014), to name but a few. We leave further work on all of these issues for future research.

Acknowledgments

The authors thank the Editor (Kay Giesecke) and two anonymous referees for their constructive comments and suggestions. The authors also thank Daniele Bianchi (discussant), Charles Clarke (discussant), Nicola Fusari (discussant), Clifton Green, Amit Goyal, Hao Jiang, Yuan Liao, Markus Pelger, Dave Rapach (discussant), Junbo Wang (discussant), Peixuan Yuan, and Guofu Zhou, along with participants at the NBER Big Data and High-Performance Computing for Financial Economics Conference, the Western Finance Association Annual Meeting, the China International Conference in Finance Annual Meeting, the Shanghai Forum by Fudan University, the Australasian Finance and Banking Conference, the Machine Learning and Financial Econometrics Workshop at Oxford-Man Institute, the American Finance Association Annual Meeting, the Midwest Finance Association Annual Meeting, the Citi Quantitative Research Conference, the European Finance Association Annual Meeting, the NBER-NSF Time Series Conference, and seminar participants at Durham University, Rutgers Business School, Renmin University, the Virtual Derivatives Workshop, the University of Rhode

Island, Shanghai University of Finance and Economics- Dishui Lake Advanced Finance Institute, Q Group Spring Seminar, and Goldman Sachs for their many helpful comments and suggestions. The authors are also grateful to the Q Group for awarding the paper the 2023 Jack Treynor Prize.

Appendix

A.1. Realized Variation and Semivariation Features

The Heterogeneous Autoregressive model proposed by Corsi (2009) has arguably emerged as *the* benchmark model for realized volatility-based forecasting. The model is designed to succinctly capture the effects of heterogeneous short-, medium-, and long-term signals with the forecasts of future volatilities based on linear combinations of lagged daily, weekly, and monthly realized volatilities. Motivated by the success of the traditional HAR model for forecasting RVs, our first feature set for forecasting the month $t + 1$ correlation for stocks i and j similarly includes the lagged daily, weekly, and monthly realized correlations for the same two stocks—say, $RC_{ij,t}^d$, $RC_{ij,t}^w$, and $RC_{ij,t}^m$. We will also refer to models based on just these three features as HAR models for short.

Moreover, motivated by Barndorff-Nielsen et al. (2010) and Patton and Sheppard (2015) and the empirical findings reported therein that HAR-type volatility forecasting models may be improved by including separate “bad,” or downside, realized volatility measures constructed from the summation of squared negative high-frequency returns (see the recent discussion in Bollerslev 2022), we also include daily, weekly, and monthly negative realized semicorrelation measures in our first feature set. In particular, following Bollerslev et al. (2020b), the annualized daily realized negative semicovariance is naturally defined by restricting the summation in (1) to the products of the negative intraday returns only:

$$RCov_{ij}^{d-} = 252 \times \sum_{k=1}^n r_{i,d-1+k/n} 1_{\{r_{i,d-1+k/n} < 0\}} r_{j,d-1+k/n} 1_{\{r_{j,d-1+k/n} < 0\}}. \quad (A.1)$$

In parallel to the construction of the weekly and monthly realized covariances, the longer horizon $RCov^-$'s may similarly be obtained by averaging the daily $RCov^-$'s defined in (A.1) over the relevant horizon, in turn allowing for the construction of the corresponding daily, weekly, and monthly semicorrelation measures—say, $RC_{ij,t}^{d-}$, $RC_{ij,t}^{w-}$, and $RC_{ij,t}^{m-}$. In total, this leaves us with six realized features, RC^d , RC^w , RC^m , RC^{d-} , RC^{w-} , and RC^{m-} , specifically designed to capture autoregressive dynamic dependencies, as well as possible asymmetric responses to signed price movements. Following the nomenclature for realized volatility models in Patton and Sheppard (2015), we will refer to correlation forecasting models based on these six features as SHAR models hereafter.

A.2. Factor-Driven Features

Numerous factor models have been proposed in the literature to account for the joint dependencies among returns through the decomposition of the total return comovements into factor-driven and residual components. Notable examples include the capital asset pricing model (CAPM) of Sharpe (1964) and Lintner (1965); the three- and five-factor

models of Fama and French (1993, 2015); the q -factor model of Hou et al. (2015); and the mispricing-factor model of Stambaugh and Yuan (2017). Several papers have further demonstrated the advantages of exploiting the factor structures in the formulation and estimation of covariance matrix forecasting models; see, for example, Chan et al. (1999), Hansen et al. (2014), Fan et al. (2016), and Alves et al. (2024). Intuitively, if a factor model captures the common return variation well, then the resulting factor-driven component of return comovement may be viewed as the systematic signal underlying the correlations. Consequently, incorporating these factor-driven correlations as additional features may enhance the predictive accuracy by focusing on the systematic drivers of the return comovements, while down-weighting the idiosyncratic noise.

The true factors, of course, are not known. Instead, following Fama and French (2020) and Li et al. (2025), we rely on observable characteristics as factor loadings to “back out” a set of factor-driven correlation features.³³ Specifically, assuming that the $N \times 1$ vector of returns is driven by K common factors, the realized covariance matrix for the returns may then be decomposed as $RCov = L \cdot RCov^f \cdot L' + RCov^e$, where L denotes the $N \times K$ matrix of factor loadings, $RCov^f$ denotes the $K \times K$ factor covariance matrix, and $RCov^e$ refers to an $N \times N$ residual covariance matrix. The off-diagonal elements in $RCov^e$ account for the return covariation that is not explained by the common factors. Correspondingly, the factor-driven covariance matrix may thus be expressed as $L \cdot RCov^f \cdot L' + \text{Diag}(RCov^e)$. The diagonal elements of that matrix naturally coincide with the diagonal elements of $RCov$. To obtain the off-diagonal elements, we substitute $(L'L)^{-1}L' \cdot RCov \cdot L(L'L)^{-1}$ in place of $RCov^f$, resulting in the factor-driven covariance matrix estimate:

$$RCov^F = L(L'L)^{-1}L' \cdot RCov \cdot L(L'L)^{-1}L' + \text{Diag}(RCov^e). \quad (\text{A.2})$$

This factor-driven covariance matrix at a given horizon may therefore be estimated from the observable characteristic matrix L and the realized covariance matrix $RCov$ at that same horizon.³⁴

In the empirical results discussed below, we rely on a loading matrix L comprising 15 representative anomalies and firm characteristics, including the 11 mispricing anomalies from Stambaugh et al. (2012) together with the usual CAPM Beta, Size, Book-to-Market, and Reversal measures.³⁵ Because the anomalies vary quite substantially in terms of their sample means and standard deviations, we follow Gu et al. (2020) and rank-transform each anomaly, except for the CAPM Beta, into the $[-1, 1]$ interval. Having estimated the daily, weekly, and monthly $RCov^F$ covariance matrices, we finally convert each of them into correlation matrices. All said and done, this leaves us with three additional daily, weekly, and monthly factor-driven realized correlation features, denoted by FRC^d , FRC^w , and FRC^m , respectively. We will refer to forecasting models based on the previous six realized features and these three additional factor-driven features as SHAR-F models.

A.3. Exponentially Weighted Realized Features

Our third and final feature set is directly motivated by the Heterogeneous Exponential Realized Volatility (HExpRV)

model proposed by Bollerslev et al. (2018b). This model provides a particularly simple framework for flexibly incorporating longer-run dynamic dependencies into realized-volatility-based forecasting by exploiting the use of exponentially weighted moving averages of lagged daily realized volatilities. Importantly, our use of exponentially weighted features is also reminiscent of the smooth distributed lag structures popularized in the MIDAS (Mixed Data Sampling) literature, as pioneered by Ghysels et al. (2006, 2007). Whereas the MIDAS approach typically parameterizes lag weights using beta polynomials, the HExpRV model instead relies on pre-specified or estimated exponential weighting schemes. This connection is also further discussed and empirically explored in Bollerslev et al. (2018b), and we build directly on these ideas to define a new set of exponentially smoothed correlation features as flexible alternatives to the standard HAR-type lag specifications.

Analogously defined exponential-weighted negative semi-variance, covariance, and negative semicovariance measures may readily be constructed as:

$$\begin{aligned} \text{ExpRV}_{i,\tau}^{-\text{CoM}(\lambda)} &= \sum_{k=1}^{500} \frac{e^{-k\lambda}}{e^{-\lambda} + e^{-2\lambda} + \dots + e^{-500\lambda}} \text{RV}_{i,\tau-k+1}^d, \\ \text{ExpRCov}_{ij,\tau}^{\text{CoM}(\lambda)} &= \sum_{k=1}^{500} \frac{e^{-k\lambda}}{e^{-\lambda} + e^{-2\lambda} + \dots + e^{-500\lambda}} \text{RCov}_{ij,\tau-k+1}^d, \\ \text{ExpRCov}_{ij,\tau}^{-\text{CoM}(\lambda)} &= \sum_{k=1}^{500} \frac{e^{-k\lambda}}{e^{-\lambda} + e^{-2\lambda} + \dots + e^{-500\lambda}} \text{RCov}_{ij,\tau-k+1}^d, \quad (\text{A.3}) \end{aligned}$$

where λ defines the decay rate of the weights, and $\text{CoM}(\lambda)$ refers to the corresponding center-of-mass formally given by $\text{CoM}(\lambda) = e^{-\lambda} / (1 - e^{-\lambda})$.³⁶ The center-of-mass effectively captures the “average” horizon of the lagged daily measures on which a given *Exp* measure is based. We construct *ExpRV*, *ExpRV*[−], *ExpRCov*, and *ExpRCov*[−] measures with center-of-mass equal to 1, 5, 21, and 63 trading days, corresponding to half-lives of one day (d), one week (w), one month (m), and one quarter (q), respectively. We then convert the resulting variance and covariance terms defined by *ExpRV* and *ExpRCov*, and *ExpRV*[−] and *ExpRCov*[−], respectively, into their respective exponential-weighted realized correlation and negative semicorrelation features, denoted by *ExpRC* ^{d} , *ExpRC* ^{w} , *ExpRC* ^{m} , *ExpRC* ^{q} , *ExpRC* ^{d −}, *ExpRC* ^{w −}, *ExpRC* ^{m −}, and *ExpRC* ^{q −} in what follows.

Additionally, in light of the strong within-sector asset comovements previously documented in the literature (see, e.g., Fan et al. 2016, Herskovic et al. 2016), we also consider a series of sector-specific exponentially weighted realized features. We begin by dividing all of the stocks in our sample into sectors.³⁷ For each of the eight exponential features defined above and each sector Sc , we then compute the average value of the features across all of the stock pairs within a particular sector, denoting the resulting sector-specific exponentially weighted realized features by *ExpScRC* ^{d} , *ExpScRC* ^{w} , *ExpScRC* ^{m} , *ExpScRC* ^{q} , *ExpScRC* ^{d −}, *ExpScRC* ^{w −}, *ExpScRC* ^{m −}, and *ExpScRC* ^{q −}, respectively. If a pair of stocks belong to different sectors, their sector-specific features are all set to zero. This, in turn, leaves us with 16 additional features. We will denote SHAR-F models that include all of these additional features as SHAR-F-Exp models for short.

Endnotes

¹ See also Giglio et al. (2022) for a survey of recent methodological contributions in asset pricing involving the use of factor models and techniques adopted from machine learning.

² As further discussed below, the simple-to-implement HAR model has arguably emerged as *the* benchmark model in the burgeon realized volatility literature. Further corroborating this status, the new results reported in Section 5 also show that correlation forecasts from the HAR model significantly enhance the performance of pairs trading strategies and equity premium predictions compared to the results obtained from the use of historical sample correlations.

³ For simplicity and ease of implementation, in our main analyses, we focus on stocks with full return histories. However, in Section 6.3, we further examine the out-of-sample performance of various models using an extended sample that relaxes this full-return-history requirement.

⁴ As discussed further below, our choice of a 15-minute sampling frequency also mirrors that of many other recent studies; see, for example, Fan et al. (2016) and Bollerslev et al. (2020b).

⁵ The volatility signature plot (defined by Andersen et al. 2000) for the realized correlations in the Online Appendix further underscores the soundness of that choice. For an early discussion of market microstructure noise and the role of sampling frequency in the context of portfolio choice based on high-frequency data, see also Bandi et al. (2008).

⁶ The CCC-GARCH model of Bollerslev (1990) and the DCC model of Engle (2002) also both rely on this same decomposition for forecasting the covariance matrix $RCov_t$ through the separate modeling and forecasting of RV_t and RC_t .

⁷ To keep our analyses simple and easy to implement, we purposely do not include any implied correlation features from options. It is certainly possible that the use of option prices could further enhance the results discussed below. However, the limited coverage and lack of liquidity for many of the individual stocks included in our sample, combined with the need to properly “clean” the data, would render a simple fit-all approach like the one we propose untenable.

⁸ To avoid any numerical issues, we further bound all empirically calculated features to lie between -1 and 1 by replacing any values above 1 (below -1) with 1 (-1).

⁹ The efficiency of our estimation procedure is further demonstrated by a transfer learning exercise discussed in Section 6.3.

¹⁰ To ensure the predicted correlations are bounded between $[-1, 1]$, we also apply an “insanity filter” and replace any predictions that fall outside that interval with 1 and -1 , respectively.

¹¹ The past two decades have witnessed the exceptional growth of many high-tech firms, and the 10 largest stocks in the S&P 500 now make up around 30% of the index’s market value. Accordingly, we purposely winsorize market capitalization at 90% to control for the effect of mega firms.

¹² Additional results reported in Table A.3 in the Online Appendix further show that these improvements accrue differently across different pairs of stocks. For instance, the gains in forecast accuracy tend to be larger for smaller stocks, value stocks, and stocks with higher CAPM betas. Meanwhile, we do not explore these differences any further here, choosing instead to focus our empirical analyses and discussion on the average improvements observed across all stock pairs within our sample.

¹³ It is important to keep in mind that the variable selection achieved by LASSO is purely statistical and that the results need to be interpreted accordingly; for additional discussion along these lines, see also Freijeiro-González et al. (2022).

¹⁴ As an alternative to our deliberately engineered realized HAR features, we also experimented with the inclusion of all lagged daily

realized correlations from day $t - 2$ to day $t - 21$ in the LASSO-based estimation. The equal-weighted and value-weighted out-of-sample R^2 ’s relative to the baseline HAR model for these more elaborate models equal 6.68% and 4.70%, respectively. Notably, however, this out-performance remains inferior to that of our preferred specification with the 25 well-designed features. As further illustrated in Figure A.4 in the Online Appendix, the freely selected lag structures are also very difficult to interpret from an economic perspective.

¹⁵ Because our sample only includes a single major recession, we caution against overinterpreting these patterns. A more thorough investigation of macro-state-dependent feature selection would be an interesting direction for future research.

¹⁶ For value-weighted quintile portfolios, we further impose the restriction that none of the weights exceed 10% to ensure that the portfolio is diversified.

¹⁷ The pairs trading strategy in Chen et al. (2019) is formed based on a different stock universe over the extended period from January 1931 to December 2007. However, from Figure 1 in their paper, the value-weighted strategy also performed poorly toward the end of their sample period, with negative annual returns in six out of nine years between 1999 and 2007. Their equal-weighted strategy performed better than the value-weighted counterpart, yet it still generated a negative return in 2007 at the end of their sample.

¹⁸ We estimate $Beta$ by regressing monthly stock excess returns on monthly market excess returns using a 60-month rolling window. $Size$ is the natural logarithm of the market value of equity, estimated by the product of the closing price and the number of shares outstanding. $Book-to-market$ ratio is the ratio of the book value of common equity to the market value of equity. $Momentum$ is estimated by the cumulative return over the past 2–12 months.

¹⁹ As shown in Figure A.2 and Table A.4 in the Online Appendix, pairs-trading strategies augmented by SHAR-F-Exp forecasts deliver performance on par with HAR-augmented strategies. The return spreads are 6.01% (equal-weighted) and 5.35% (value-weighted) per annum with $t = 1.40$ and 1.30 , respectively. The coefficients on $RetDiff$ are 0.08 and 0.11, both insignificantly different from zero, indicating that the observed gains primarily stem from LASSO’s variable-selection and regularization properties.

²⁰ For a survey of the latest literature on market return predictability, see, for example, Rapach and Zhou (2022).

²¹ To prevent excessive influence from mega-cap stocks, we winsorize firm-level market capitalizations at the 90th percentile each month.

²² All macroeconomic predictors are downloaded from Amit Goyal’s personal website. dp is the dividend-price ratio computed as the difference between the log of 12-month moving sums of dividends and the log of prices of the S&P 500 index. ep is the earnings-price ratio calculated as the difference between the log of 12-month moving sums of earnings and the log of prices of the S&P 500 index. bm is the book-to-market ratio defined as the ratio of book value to market value for the Dow Jones Industrial Average. The net equity expansion $ntis$ is a ratio of 12-month moving sums of net issues by NYSE-listed stocks divided by the total end-of-year market capitalization of NYSE stocks. tbl is the three-month Treasury Bill rate. tms is the difference between the long-term yield on government bonds and the Treasury bill. dfy is the difference between BAA- and AAA-rated corporate bond yields. $svar$ is the sum of squared daily returns on the S&P 500.

²³ We also evaluate the risk-targeting performance for 15 additional anomalies pertaining to various aspects of firms’ operation and market performance, as further detailed in Table A.2 in the Online Appendix. The ratios for these additional strategies, displayed in Figure A.3 in the Online Appendix, exhibit very similar patterns.

²⁴ For each strategy, we have 155-month observations of the risk-targeting ratio. We perform 1,000 iterations of the bootstrap, where for each iteration, we draw 155 ratios from the original data with replacement. For each bootstrapped sample, we then compute the average risk-target ratio, using the 2.5th and 97.5th percentiles of the resulting 1,000 average risk-targeted ratios as our bootstrapped confidence interval.

²⁵ Interestingly, Jagannathan and Ma (2003) also find that GMV portfolios often achieve higher out-of-sample Sharpe ratios than mean-variance optimized tangent portfolios.

²⁶ More advanced shrinkage procedures, such as those proposed by Ledoit and Wolf (2003, 2004a, 2004b), could alternatively be employed and may lead to further improvements.

²⁷ It is worth noting that even though we do not impose any constraint on the weights (as done in, e.g., Chan et al. 1999), neither of the two models generates especially extreme weights, as evidenced by the minimum portfolio weights for both the HAR- and LASSO-based predictions never falling below -10% , and the maximum portfolio weights being around 30% for both models across all stocks and months in the sample.

²⁸ Concretely, the standard deviations of the HAR-based GMV portfolios exceed those of the LASSO-based GMV portfolios for 77% of the months in our sample. Similar patterns are observed for the SHAR-F-Exp-based portfolios, which exhibit an average portfolio standard deviation of 10.36% , with most data points lying above the 45-degree line in the corresponding scatter plot of monthly risks.

²⁹ A number of other firm linkage measures have also been explored in the literature, such as stocks with similar value-growth labeling (Boyer 2011) and firms with similar retail concentration (Kumar and Lee 2006).

³⁰ There is a long history on the use of neural networks for forecasting equity volatility, as exemplified by the early work by Donaldson and Kamstra (1997).

³¹ For the Principal Component Regressions, to help increase the computational speed and prevent overfitting, we restrict the number of components used in the regressions to be no more than 20. For the Neural Network, we report the results based on a two-hidden-layer feed-forward network with 4 and 2 neurons. However, in unreported results, we also considered a three-hidden-layer network with 8, 4, and 2 neurons and a four-hidden-layer network with 16, 8, 4, and 2 neurons.

³² The inclusion of manually selected interacted terms in the largest feature set further facilitates a fair comparison among the linear models (i.e., LASSO, Ridge, ENet, and PCR) and the nonlinear models (i.e., FNN) and helps pin down the contribution of the interactive effects.

³³ Clarke and Linn (2024) offer empirical evidence suggesting that characteristics can proxy for systematic risk exposure.

³⁴ A similar approach has also previously been used by Fan et al. (2016) in their estimation of the factor covariance matrices from high-frequency factors formed by firm characteristic sorts.

³⁵ Table A.1 in the Online Appendix provides further details and summary statistics for each of the 15 anomaly variables.

³⁶ Conversely, for a given center-of-mass, the corresponding decay rate equals $\lambda = \log(1 + 1/\text{CoM})$.

³⁷ Our sector classifications are based on the first two digits of the stocks' Global Industry Classification Standard codes from Compustat. The 10 sectors include: Energy (10), Materials (15), Industrials (20), Consumer Discretionary (25), Consumer Staples (30), Health Care (35), Financials (40), Information Technology (45), Communication Services (50), and Utilities (55).

References

- Ait-Sahalia Y, Fan J, Xiu D (2010) High-frequency covariance estimates with noisy and asynchronous financial data. *J. Amer. Statist. Assoc.* 105(492):1504–1517.
- Alves RP, Brito DSD, Medeiros MC, Ribeiro RM (2024) Forecasting large realized covariance matrices: The benefits of factor models and shrinkage. *J. Financial Econometrics* 22(3):696–742.
- Andersen TG, Bollerslev T, Diebold FX, Labys P (2000) Great realizations. *Risk* 13:105–108.
- Andersen TG, Bollerslev T, Diebold FX, Labys P (2003) Modeling and forecasting realized volatility. *Econometrica* 71(2):579–625.
- Antón M, Polk C (2014) Connected stocks. *J. Finance* 69(3):1099–1127.
- Appel IR, Gormley TA, Keim DB (2016) Passive investors, not passive owners. *J. Financial Econom.* 121(1):111–141.
- Audrino F, Knaus SD (2016) Lassoing the HAR model: A model selection perspective on realized volatility dynamics. *Econom. Rev.* 35(8–10):1485–1521.
- Audrino F, Trojani F (2011) A general multivariate threshold GARCH model with dynamic conditional correlations. *J. Bus. Econom. Statist.* 29(1):138–149.
- Back KE (2010) *Asset Pricing and Portfolio Choice Theory* (Oxford University Press, Oxford, UK).
- Bali TG, Beckmeyer H, Mörke M, Weigert F (2023) Option return predictability with machine learning and big data. *Rev. Financial Stud.* 36(9):3548–3602.
- Bali TG, Goyal A, Huang D, Jiang F, Wen Q (2022) Predicting corporate bond returns: Merton meets machine learning. Working paper, Georgetown University, Washington, DC.
- Bandi FM, Russell JR, Zhu Y (2008) Using high-frequency data in dynamic portfolio choice. *Econom. Rev.* 27(1–3):163–198.
- Barberis N, Shleifer A, Wurgler J (2005) Comovement. *J. Financial Econom.* 75(2):283–317.
- Barndorff-Nielsen OE, Shephard N (2004) Econometric analysis of realized covariation: High-frequency based covariance, regression, and correlation in financial economics. *Econometrica* 72(3):885–925.
- Barndorff-Nielsen OE, Kinnebrock S, Shephard N (2010) Measuring downside risk: Realised semivariance. Bollerslev T, Russell J, Watson M, eds. *Volatility and Time Series Econometrics: Essays in Honor of Robert F. Engle* (Oxford University Press, Oxford, UK), 117–136.
- Bollerslev T (1990) Modelling the coherence in short-run nominal exchange rates: A multivariate generalized ARCH model. *Rev. Econom. Statist.* 72(3):498–505.
- Bollerslev T (2022) Realized semi(co)variation: Signs that all volatilities are not created equal. *J. Financial Econometrics* 20(2):219–252.
- Bollerslev T, Meddahi N, Nyawa S (2019) High-dimensional multivariate realized volatility estimation. *J. Econom.* 212(1):116–136.
- Bollerslev T, Patton AJ, Quaadvlieg R (2018a) Modeling and forecasting (un)reliable realized covariances for more reliable financial decisions. *J. Econom.* 207(1):71–91.
- Bollerslev T, Patton AJ, Quaadvlieg R (2020a) Multivariate leverage effects and realized semicovariance GARCH models. *J. Econom.* 217(2):411–430.
- Bollerslev T, Patton AJ, Zhang H (2022a) Equity clusters through the lens of realized semicorrelations. *Econom. Lett.* 211:110245.
- Bollerslev T, Hood B, Huss J, Pedersen LH (2018b) Risk everywhere: Modeling and managing volatility. *Rev. Financial Stud.* 31(7):2729–2773.
- Bollerslev T, Li J, Patton AJ, Quaadvlieg R (2020b) Realized semicovariances. *Econometrica* 88(4):1515–1551.
- Bollerslev T, Medeiros MC, Patton AJ, Quaadvlieg R (2022b) From zero to hero: Realized partial (co)variances. *J. Econom.* 231(2):348–360.
- Bondarenko O, Bernard C (2024) Option-implied dependence and correlation risk premium. *J. Financial Quant. Anal.* 59(7):3139–3189.

- Boyer BH (2011) Style-related comovement: Fundamentals or labels. *J. Finance* 66(1):307–332.
- Bucci A (2020) Realized volatility forecasting with neural networks. *J. Financial Econometrics* 18(3):502–531.
- Buraschi A, Kosowski R, Trojani F (2014) When there is no place to hide: Correlation risk and the cross-section of hedge fund returns. *Rev. Financial Stud.* 27(2):581–616.
- Buss A, Vilkov G (2012) Measuring equity risk with option-implied correlations. *Rev. Financial Stud.* 25(10):3113–3140.
- Cappiello L, Engle RF, Sheppard K (2006) Asymmetric dynamics in the correlations of global equity and bond returns. *J. Financial Econometrics* 4(4):537–572.
- Chan LKC, Karceski J, Lakonishok J (1999) On portfolio optimization: Forecasting covariances and choosing the risk model. *Rev. Financial Stud.* 12(5):937–974.
- Chen AY, Zimmermann T (2022) Open source cross-sectional asset pricing. *Critical Finance Rev.* 27:207–264.
- Chen L, Pelger M, Zhu J (2024) Deep learning in asset pricing. *Management Sci.* 70(2):714–750.
- Chen H, Chen S, Chen Z, Li F (2019) Empirical investigation of an equity pairs trading strategy. *Management Sci.* 65(1):370–389.
- Christensen K, Siggaard M, Veliyev B (2023) A machine learning approach to volatility forecasting. *J. Financial Econometrics* 21(5):1680–1727.
- Clarke C, Linn M (2024) Characteristics and the cross-section of covariances. Working paper, North Carolina State University, Raleigh.
- Cong LW, Feng G, He J, He X (2025) Growing the efficient frontier on panel trees. *J. Financial Econ.* 167:104024.
- Cong LW, Feng G, He J, Wang Y (2024) Mosaics of predictability. Working paper, Nanyang Technological University, Singapore.
- Corsi F (2009) A simple approximate long-memory model of realized volatility. *J. Financial Econometrics* 7(2):174–196.
- Cosemans M, Frehen R, Schotman PC, Bauer R (2016) Estimating security betas using prior information based on firm fundamentals. *Rev. Financial Stud.* 29(4):1072–1112.
- Diebold FX, Mariano RS (1995) Comparing predictive accuracy. *J. Bus. Econom. Statist.* 13(3):253–263.
- Donaldson RG, Kamstra M (1997) An artificial neural network-GARCH model for international stock return volatility. *J. Empirical Finance* 4(1):17–46.
- Dong X, Li Y, Rapach D, Zhou G (2022) Anomalies and expected market return. *J. Finance* 77(1):639–681.
- Driessen J, Maenhout PJ, Vilkov G (2009) The price of correlation risk: Evidence from equity options. *J. Finance* 64(3):1377–1406.
- Driessen J, Maenhout PJ, Vilkov G (2013) Option-implied correlations and the price of correlation risk. Working paper, Tilburg University, Tilburg, Ethiopia.
- Engle R (2002) Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *J. Bus. Econom. Statist.* 20(3):339–550.
- Engle RF, Ledoit O, Wolf M (2019) Large dynamic covariance matrices. *J. Bus. Econom. Statist.* 37(2):363–375.
- Eppe TW (1979) Comovements in stock prices in the very short run. *J. Amer. Statist. Assoc.* 74(366):291–298.
- Fama EF, French KR (1993) Common risk factors in the returns on stocks and bonds. *J. Financial Econ.* 33(1):3–56.
- Fama EF, French KR (2015) A five-factor asset pricing model. *J. Financial Econ.* 116(1):1–22.
- Fama EF, French KR (2020) Comparing cross-section and time-series factor models. *Rev. Financial Stud.* 33(5):1891–1926.
- Fan J, Furger A, Xiu D (2016) Incorporating global industrial classification standard into portfolio allocation: A simple factor-based large covariance matrix estimator with high-frequency data. *J. Bus. Econom. Statist.* 34(4):489–503.
- Fleming J, Kirby C, Ostdiek B (2003) The economic value of volatility timing using “realized” volatility. *J. Financial Econ.* 67(3):473–509.
- Freijeiro-González L, Febrero-Bande M, González-Manteiga W (2022) A critical review of LASSO and its derivatives for variable selection under dependence among covariates. *Internat. Statist. Rev.* 90(1):118–145.
- Ghysels E, Santa-Clara P, Valkanov R (2006) Predicting volatility: Getting the most out of return data sampled at different frequencies. *J. Econom.* 131(1–2):59–96.
- Ghysels E, Sinko A, Valkanov R (2007) MIDAS regressions: Further results and new directions. *Econometric Rev.* 26(1):53–90.
- Giglio S, Kelly B, Xiu D (2022) Factor models, machine learning, and asset pricing. *Annual Rev. Financial Econ.* 14:337–368.
- Green J, Hand JRM, Zhang XF (2013) The superview of return predictive signals. *Rev. Accounting Stud.* 18:692–730.
- Gu S, Kelly B, Xiu D (2020) Empirical asset pricing via machine learning. *Rev. Financial Stud.* 33(5):2223–2273.
- Hameed A, Morck R, Shen J, Yeung B (2015) Information, analysts, and stock return comovement. *Rev. Financial Stud.* 28(11):3153–3187.
- Hansen PR, Lunde A (2005) A realized variance for the whole day based on intermittent high-frequency data. *J. Financial Econometrics* 3(4):525–554.
- Hansen PR, Lunde A, Voev V (2014) Realized beta GARCH: A multivariate GARCH model with realized measures of volatility. *J. Appl. Econometrics* 29(5):774–799.
- Herskovic B, Kelly B, Lustig H, Van Nieuwerburgh S (2016) The common factor in idiosyncratic volatility: Quantitative asset pricing implications. *J. Financial Econ.* 119(2):249–283.
- Hoberg G, Phillips G (2010) Product market synergies and competition in mergers and acquisitions: A text-based analysis. *Rev. Financial Stud.* 23(10):3773–3811.
- Hoberg G, Phillips G (2016) Text-based network industries and endogenous product differentiation. *J. Political Econ.* 124(5):1423–1465.
- Hou K, Xue C, Zhang L (2015) Digesting anomalies: An investment approach. *Rev. Financial Stud.* 28(3):650–705.
- Israelen RD (2016) Does common analyst coverage explain excess comovement? *J. Financial Quant. Anal.* 51(4):1193–1229.
- Jagannathan R, Ma T (2003) Risk reduction in large portfolios: Why imposing the wrong constraints helps. *J. Finance* 58(4):1651–1683.
- Johansson K, Ogut MG, Pelger M, Schmelzer T, Boyd S (2023) A simple method for predicting covariance matrices of financial returns. *Foundations Trends Econometrics* 12(4):324–407.
- Kaniel R, Lin Z, Pelger M, Nieuwerburgh S (2023) Machine-learning the skill of mutual fund managers. *J. Financial Econ.* 150(1):94–138.
- Kelly BT, Malamud S, Zhou K (2024) The virtue of complexity in return prediction. *J. Finance* 79(1):459–503.
- Krishnan C, Petkova R, Ritchken P (2009) Correlation risk. *J. Empirical Finance* 16(3):353–367.
- Kumar A, Lee CM (2006) Retail investor sentiment and return comovements. *J. Finance* 61(5):2451–2486.
- Ledoit O, Wolf M (2003) Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *J. Empirical Finance* 10(5):603–621.
- Ledoit O, Wolf M (2004a) Honey, I shrunk the sample covariance matrix. *J. Portfolio Management* 30(4):110–119.
- Ledoit O, Wolf M (2004b) A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivariate Anal.* 88(2):365–411.
- Li B, Rossi A (2025) Selecting mutual funds from the stocks they hold: A machine learning approach. Working paper, Wuhan University, Wuhan, China.
- Li SZ, Tang Y (2025) Automated volatility forecasting. *Management Sci.* 71(7):6248–6274.
- Li SZ, Yuan P, Zhou G (2025) Systematic momentum: A new class of price patterns. *Management Sci.*, ePub ahead of print November 25, <https://doi.org/10.1287/mnsc.2024.08236>.
- Lintner J (1965) The valuation of risky assets and the selection of risky investments in stock portfolios and capital budgets. *Rev. Econom. Statist.* 47(1):13–37.

- Lopez-Lira A, Roussanov NL (2023) Do common factors really explain the cross-section of stock returns? Working paper, University of Florida, Gainesville.
- Menzly L, Ozbas O (2010) Market segmentation and cross-predictability of returns. *J. Finance* 65(4):555–1580.
- Mueller P, Stathopoulos A, Vedolin A (2017) International correlation risk. *J. Financial Econom.* 126(2):270–299.
- Muslu V, Rebello M, Xu Y (2014) Sell-side analyst research and stock comovement. *J. Accounting Res.* 52(4):911–954.
- Noureldin D, Shephard N, Sheppard K (2012) Multivariate high-frequency-based volatility (HEAVY) models. *J. Appl. Econometrics* 27(6):907–933.
- Oh DH, Patton AJ (2016) High-dimensional copula-based distributions with mixed frequency data. *J. Econom.* 193(2):349–366.
- Parsons CA, Sabbatucci R, Titman S (2020) Geographic lead-lag effects. *Rev. Financial Stud.* 33(10):4721–4770.
- Patton AJ, Sheppard K (2015) Good volatility, bad volatility: Signed jumps and the persistence of volatility. *Rev. Econom. Statist.* 97(3):683–697.
- Pindyck RS, Rotemberg JJ (1993) The comovement of stock prices. *Quart. J. Econom.* 108(4):1073–1104.
- Pirinsky C, Wang Q (2006) Does corporate headquarters location matter for stock returns? *J. Finance* 61(4):1991–2015.
- Pollet JM, Wilson M (2010) Average correlation and stock market returns. *J. Financial Econom.* 96(3):364–380.
- Rapach DE, Zhou G (2022) Asset pricing: Time-series predictability. Banerjee A, ed. *Oxford Research Encyclopedia of Economics and Finance* (Oxford University Press, Oxford, UK).
- Rapach DE, Strauss JK, Zhou G (2013) International stock return predictability: What is the role of the United States? *J. Finance* 68(4):1633–1662.
- Roll R (1984) A simple implicit measure of the effective bid-ask spread in an efficient market. *J. Finance* 39(4):1127–1139.
- Sharpe WF (1964) Capital asset prices: A theory of market equilibrium under conditions of risk. *J. Finance* 19(3):425–442.
- Stambaugh RF, Yuan Y (2017) Mispricing factors. *Rev. Financial Stud.* 30(4):1270–1315.
- Stambaugh RF, Yu J, Yuan Y (2012) The short of it: Investor sentiment and anomalies. *J. Financial Econom.* 104(2):288–302.
- Tibshirani R (1996) Regression shrinkage and selection via the LASSO. *J. Roy. Statist. Soc. Ser. B* 58(1):267–288.
- Timmermann A (2006) Forecast combinations. Elliott G, Granger CWJ, Timmermann A, eds. *Handbook of Economic Forecasting*, vol. 1 (Elsevier, Amsterdam), 135–196.
- Tse YK, Tsui AKC (2002) A multivariate generalized autoregressive conditional heteroscedasticity model with time-varying correlations. *J. Bus. Econom. Statist.* 20(3):351–362.
- Welch I, Goyal A (2008) A comprehensive look at the empirical performance of equity premium prediction. *Rev. Financial Stud.* 21(4):1455–1508.